From one bit to a fitted polynomial: engineered spectral filters for track reconstruction in a noisy LHCb VELO toy model

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Abstract

Track reconstruction in the LHCb vertex detector can be written as a linear system $A\mathbf{x} = \mathbf{b}$ over track *segments*, whose solution scores each segment by how strongly its neighbours reinforce it. Two earlier papers solved this system with quantum linear algebra: first with the HHL algorithm, and then far more cheaply with a one-bit variant, the one-bit quantum filter (1BQF), which replaces the full inversion by a single cosine filter of the Hamiltonian with one spectral zero. In this work I develop that idea further. The 1BQF is the degree-one member of the family of polynomial spectral transforms provided by the quantum singular value transformation (QSVT), and once the problem is viewed this way the design question changes: it is no longer a question of which algorithm inverts A, but of which function of λ separates true from false segments. On clean events the answer is a four-line comb pinned to the eigenvalues of a five-hit track. Read, as every solver in this work is, at the efficiency-first working point (the lowest threshold that keeps 99% of the true segments), it holds the segment false rate below the classical solver's up to $T \approx 400$ tracks at the wide fixed 2 mrad acceptance; at T=1000, where tangles of false segments have entered the clean-event design's passes, the same degree-40 polynomial with its coefficients refit to the events reads a 2.8% false rate where the classical solver reads 10.0%, and at the formula acceptance the same multiplicity is solved outright (99.98% efficiency at a false rate below 0.1%). On noisy events the answer must be measured: the false weight piles up on the eigenvalue lines of specific two- and three-segment motifs, and a polynomial fitted with roots at the measured pile-up beats the classical baseline at every attainable efficiency below a sharply defined ceiling. That ceiling has a structural cause: a broken track's two-segment fragment and an accidental false pair are the same object as far as A is concerned — two segments sharing one hit, hence the same sub-block, hence the same eigenvalues and the same amplitudes — so on that Hamiltonian no function of λ can separate them, and their population is what caps segment efficiency near 0.98. I also show that the two natural Hamiltonian modifications inherited from the classical literature can only be judged together with the response applied to them. The Denby–Peterson occupancy term destroys the one-bit filter, whose single zero cannot follow the spectrum it creates; but paired with a response refitted to that spectrum it becomes the strongest configuration in the study: at the matched 99%-efficiency point (T=200) it cuts the false rate thirtyfold at moderate noise (0.31 to 0.010), and at heavy noise by a fifth to a half — both figures depend on the polynomial degree, the moderate one reading seventeenfold at degree 40 against thirtyfold at degree 80 — and, at small coupling, it breaks the twin degeneracy that sets the ceiling, which is therefore a property of the base Hamiltonian and not of the problem. The bifurcation (fork) term is likewise a regression under the clean-event comb applied without refitting, and likewise a win once the response is

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refit to the spectrum it creates, but lower on the efficiency axis: below the ceiling it wins, and at the strict 99% point both systems flood alike. The result is a two-part design rule: the operator shapes the spectrum, and the fitted response separates it; neither knob can be evaluated without the other.

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1 Introduction

Charged-particle track reconstruction is, at heart, a combinatorial matching problem: a detector hands us a cloud of hits, and we must decide which hits belong together. At the High-Luminosity LHC this cloud becomes dense enough that the matching cost dominates real-time reconstruction budgets, and the tracking community has responded by testing essentially every alternative computing platform: GPUs, FPGAs, neural networks, and, in the line of work continued here, quantum computers.

The lineage of this work is short and worth stating precisely. Nicotra et al. [1] showed that track reconstruction in a simplified model of the LHCb vertex detector (VELO) can be phrased as a linear system $A\mathbf{x} = \mathbf{b}$ over track segments and solved with the Harrow–Hassidim–Lloyd algorithm [2]; a restricted-phase-estimation variant of that construction was subsequently reported in ref. [4]. The follow-up paper [3] observed that the full inversion is overkill. The interesting information is which segments are active, not their precise weights, so a single-ancilla circuit, the one-bit quantum filter (1BQF), suffices. The 1BQF applies one cosine function of the Hamiltonian, placed so that its zero lands exactly on the eigenvalue shared by every isolated fake segment, and reads out the survivors. It is cheap, it suits near-term hardware, and it was validated against device-calibrated noise models of trapped-ion and superconducting processors; running it on the devices themselves remains to be done.

This work asks the question that follows naturally from the 1BQF. A cosine with one zero is a polynomial filter of degree one[1]. The quantum singular value transformation (QSVT) [7, 10] tells us that a quantum computer can apply essentially any bounded polynomial of the Hamiltonian at a cost linear in its degree. So, what could we do with more than one zero? Which polynomial should we choose? And what happens to that choice when the Hamiltonian itself changes, because the detector is noisy or because we have added constraint terms to help it? In my opinion this last question carries most of the physics, and most of this work is spent answering it.

I answer these questions on the same toy model as the two predecessor papers, extended with

[1] Explicitly, from the circuit of section 3: the interferometer keeps the branch $\frac{1}{2}\left(I + e^{iAt}\right) = e^{iAt/2}\cos(At/2)$, and the leading unitary drops out of the read-out of $|\langle j|\psi\rangle|$, so the device applies $\cos(At/2)$ exactly. In the variable $u = \cos(\lambda t/2)$ that response *is* the polynomial $p(u) = u$: degree one, and therefore one root, at $u = 0$, i.e. at $\lambda = \pi/t = s$. Each further application of the same controlled evolution buys one more degree and with it one more root; section 4 gives the general construction, in which d calls of a fixed walk operator realise any bounded polynomial of degree d.

realistic noise. The answers can be summarised as a single working method, which I will refer to as fitting the response to the spectrum:

1. On clean events the true-track spectrum is *discrete*: detector geometry pins every true track to the same five-hit chain, whose four eigenvalues are known in closed form. A comb of narrow passes at those four lines, and nothing else, separates true from false essentially perfectly at low and moderate multiplicity; at the densities I can simulate ($T = 1000$ tracks, four million segments) and the wide fixed 2 mrad acceptance, where the classical solver's false rate at 99% efficiency has climbed to 10.0%, the comb's coefficients must be refit to the events, and then read 2.8%.

2. On noisy events the false population moves: it concentrates on the eigenvalue lines of specific small *motifs* (hit-sharing pairs and triples), which sit off the true lines but inside any naive pass window. The correct response cannot be written down a priori; it must be fitted to the measured pile-up. Once it is, the quantum false rate drops below the classical one at every efficiency below a ceiling that I can identify, count, and explain segment by segment.

3. Hamiltonian engineering and response engineering are coupled, and neither knob can be evaluated without the other. Both Denby–Peterson penalties read as regressions under the clean-event comb applied without refitting its coefficients, and both are rehabilitated once those coefficients are refit to the spectrum they produce — but not equally, and not at the same place on the efficiency axis. On the one-bit filter the two act in opposite directions: the fork term helps it, while the occupancy term destroys it outright, its single zero having no way to follow the co-hit coupling the term creates. The occupancy term is the sharpest case in the study: with the polynomial's coefficients refit it cuts the false rate at the matched 99%-efficiency point thirtyfold at moderate noise and by a fifth to a half at heavy noise (T=200), both of them degree-dependent — at degree 40 the moderate figure is seventeenfold, at degree 80 thirtyfold — and at small coupling it breaks the twin degeneracy that caps segment efficiency near 0.98 on the base Hamiltonian. The fork term wins too, but only below that cap. Changing the operator moves the activations, and moving the activations changes the polynomial we have to fit.

This work is organised as follows. Section 2 builds the toy model and the segment Hamiltonian from zero: the event generator and the noise triple it takes, the exact activation levels that make everything else quantitative, and the conventions every later number obeys — the efficiency-first working point, and the definitions of segment efficiency and segment false rate. Section 3 reads the 1BQF as a spectral filter, deriving what its single zero removes, what it leaves behind, and how it reshapes the true band. Section 4 develops the QSVT machinery (block encoding, qubitization, and the linear-combination-of-Chebyshev construction) at the level of detail needed to count qubits and gates.

Section 5 designs and evaluates the comb on clean events, including its resource scaling and the twin degeneracy that limits every response in the family. Section 6 moves to the heavy operating point and measures the structure the noise creates: no giant tangle, but a population of small false motifs. Section 7 dissects the occupancy term's collision with the notch. Section 8 introduces the fork term and the design rule that every operator change demands a response refit. Section 9 fits the polynomial to the measured false pile-up and presents the main comparison, including the refitted occupancy result. Section 10 then crosses the two axes systematically — every Hamiltonian against every polynomial degree, with the coefficients refit at each — and reports the loss budget, the density scan that carries every solver to $T = 1000$, and the two-axis rule the study reduces to. Section 11 collects the limitations: that the occupancy win does not survive density, the realizability of a refitted response, the single-response question, Trotterization, the hardware-resource comparison of HHL, the 1BQF and the QSVT — measured for the latter two, a structure-level estimate for HHL — with the error-correction requirements each implies, a

hardware-native block encoding priced by detector hits rather than matrix entries, the success probability and readout cost, what the classical simulability of the response family does and does not concede, and what a real detector geometry changes. Section 12 concludes.

A remark on style before we begin. This work is deliberately verbose. The predecessor papers compressed the segment-Hamiltonian formalism to a few paragraphs; here I want every level, every eigenvalue, and every convention on the table, because the physics of why filters succeed and fail lives in exactly those details.

2 The toy model and the segment Hamiltonian

2.1 Events

The toy detector is the standard simplified VELO of refs. [1, 3]: parallel planes perpendicular to the beam axis (z), each recording a two-dimensional point where a charged particle crosses it. Every parameter of the generator is fixed for the whole of this work, and I list them here so that any event in it can be reproduced: five planes at $z = 33, 66, 99, 132, 165$ mm, so the inter-plane spacing is $\Delta z = 33$ mm, each plane square of half-width 40 mm in x and y; T tracks per event from a common primary vertex, itself drawn on the beam axis with $\sigma_z = 1$ mm and no transverse spread; and generator angles bounded by $\varphi_{\max} = \theta_{\max} = 0.2$ rad, so the tracks fill a cone of half-opening 0.2. Each track is propagated as a straight line and leaves one hit per plane.

Detector effects enter as controlled perturbations, and I will always state the triple in force: multiple-scattering kinks of width σ_{scatt} applied at each plane crossing, Gaussian hit-position smearing of width σ_{res}, and a hit-drop (inefficiency) probability applied hit by hit. Three configurations are used in this work and no others:

	σ_{scatt}	σ_{res}	drop	ε
clean	10^{-4}	0	0	0.42 mrad
moderate	10^{-4}	10 μm	1%	3.18 mrad
heavy	10^{-4}	20 μm	1%	6.31 mrad

where ε is the angular acceptance that eq. (6) below derives from each triple. Clean events carry section 5; the heavy configuration is the operating point from section 6 onward; the moderate one is the intermediate rung used wherever sections 9 and 10 scan across noise.

A *segment* is an ordered pair of hits on adjacent planes. With five planes and T hits per plane, each of the four inter-plane gaps offers T^2 candidate segments, so

$$n_{\text{seg}} = 4T^2, \qquad n_{\text{true}} = 4T, \qquad n_{\text{false}} = 4T(T-1), \tag{1}$$

where n_{true} counts the segments that connect two hits of the same generated track. The problem is a needle in a haystack, and the ratio $n_{\text{true}}/n_{\text{seg}} = 1/T$ says how deep the hay is: at $T = 1000$ only one segment in a thousand is real, and the false population grows quadratically while the true one grows linearly. Any segment classifier, classical or quantum, fights this composition before it fights anything else.

2.2 The Hamiltonian

Following Denby and Peterson's neural formulation of track finding [5, 6] as adapted by refs. [1, 3], we encode an event in a quadratic energy over segment activations $x_i \in \mathbb{R}$,

$$E(\mathbf{x}) = \tfrac{1}{2}\,\mathbf{x}^{\top} A\,\mathbf{x} - \delta\,\mathbf{1}^{\top}\mathbf{x}, \qquad A = (\gamma + \delta)\,I - C, \tag{2}$$

where C is the 0/1 *compatibility adjacency*: $C_{ij} = 1$ exactly when segments i and j share a hit as head-to-tail continuations (the end hit of one is the start hit of the other) and their mutual

angle is below an acceptance ε. The constants are a self-penalty γ (discouraging indiscriminate activation) and a bias δ (rewarding activation); throughout the main text $\gamma = 3$, $\delta = 1$, so the constant diagonal is $s \equiv \gamma + \delta = 4$. Minimising eq. (2) gives the linear system

$$A\,\mathbf{x} = \delta\,\mathbf{1}, \tag{3}$$

which I solve classically with MINRES. A useful picture is a voting market: each segment starts with the same bias-funded credit δ, and the off-diagonal couplings let aligned neighbours reinforce one another. A segment embedded in a long, straight chain of compatible neighbours accumulates support; a segment with no compatible continuation is left with only its own bias against its own penalty.

Whether that system is well posed at all is something the two constants buy, and it is worth being exact about how. Since A's diagonal is the constant s and C is a zero-diagonal 0/1 adjacency, the two spectra are related not by a bound but by an identity, $\lambda(A) = s - \lambda(C)$, so

$$\lambda_{\min}(A) = (\gamma + \delta) - \lambda_{\max}(C), \tag{4}$$

and A is positive definite exactly when $\lambda_{\max}(C) < \gamma + \delta$. Gershgorin applied to C turns that into something checkable without solving anything: $\lambda_{\max}(C) \le d_{\max}$, the largest number of compatible neighbours any segment has, so $\gamma + \delta > d_{\max}$ is sufficient. Positive definiteness is therefore not a property one hopes for but one the regularisation controls: on any event, raising γ past $\lambda_{\max}(C) - \delta$ restores it.

The sufficient condition is loose, and the honest statement is that sparsity, not the bound, is what keeps the benchmark solvable. At $\varepsilon = 2\,\mathrm{mrad}$ and $\gamma = 3$ some segment already reaches degree $4 = s$ by $T \approx 400$, so the Gershgorin floor $s - d_{\max}$ collapses to zero there, while the measured $\lambda_{\min}(A)$ is still 0.8–2.4 out to $T = 1000$ (figure 1). Where it does bite is the wide-acceptance corner. At the heavy configuration, evaluated exactly on each connected component, $\lambda_{\min}(A)$ falls from 1.91 at $T = 100$ through 0.71 at $T = 400$ to 0.10 ± 0.26 at $T = 700$ and 0.009 at $T = 1000$: marginal, with the occasional negative mode already appearing at $T = 700$. Push the resolution to $\sigma_{\mathrm{res}} = 50\,\mu\mathrm{m}$, where eq. (6) returns $\varepsilon = 15.8\,\mathrm{mrad}$, and $\lambda_{\max}(C)$ crosses $\gamma + \delta$ near $T \approx 290$: beyond that A is genuinely indefinite, and the classical solution begins to diverge on the offending cluster rather than merely being ill-conditioned. This is why the linear system is solved with MINRES rather than the conjugate gradient that a positive-definite matrix would invite: the same code path has to serve those corners and the $\gamma = 1$ operating point, where $2I - C$ is indefinite outright and conjugate gradient has no convergence guarantee at all.

The same operator is worth writing in the form the predecessor papers use, because the two look quite different on the page. Refs. [1, 3, 4] build the energy term by term in the Denby–Peterson manner, over activations S_i that begin binary and are then relaxed to the reals:

$$\mathcal{H}(S) = \mathcal{H}_{\mathrm{ang}}(S, \varepsilon) + \gamma\,\mathcal{H}_{\mathrm{spec}}(S) + \delta\,\mathcal{H}_{\mathrm{gap}}(S), \tag{5}$$

an angular term that rewards compatible continuations,

$$\mathcal{H}_{\mathrm{ang}} = -\tfrac{1}{2}\sum_{i,j} f(\theta_{ij}, \varepsilon)\, S_i S_j, \qquad f(\theta, \varepsilon) = \begin{cases} 1, & \cos\theta \ge 1 - \varepsilon, \\ 0, & \text{otherwise}, \end{cases}$$

plus a spectral term $\mathcal{H}_{\mathrm{spec}} = \frac{1}{2}\sum_i S_i^2$ that penalises activation and a gap term $\mathcal{H}_{\mathrm{gap}} = \frac{1}{2}\sum_i (1 - S_i)^2$ that rewards it. In refs. [3, 4] the two multipliers are called α and β; they are the γ and δ used here, as the reference implementation of ref. [3] makes explicit. Expanding the squares,

$$\mathcal{H}(\mathbf{x}) = \tfrac{1}{2}\,\mathbf{x}^{\top}\big[(\gamma + \delta)I - C\big]\mathbf{x} - \delta\,\mathbf{1}^{\top}\mathbf{x} + \tfrac{\delta}{2}\,n_{\mathrm{seg}} = E(\mathbf{x}) + \tfrac{\delta}{2}\,n_{\mathrm{seg}},$$

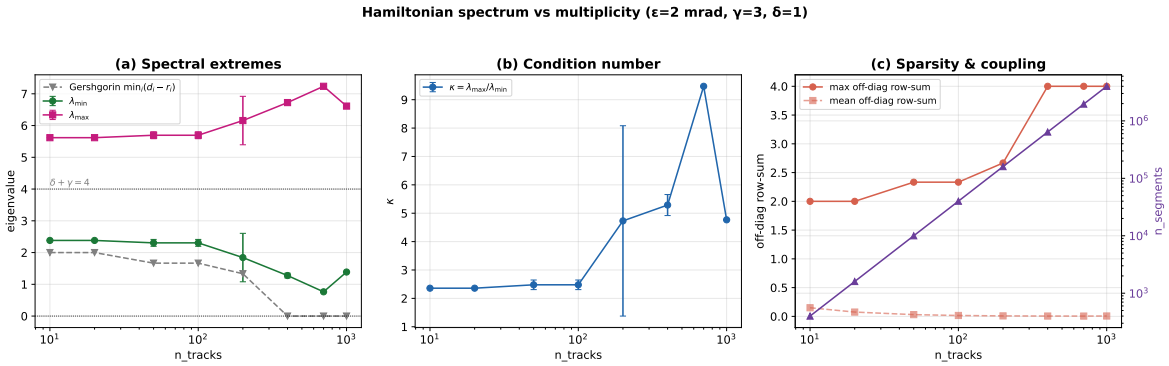


Figure 1: What keeps the system solvable, on the benchmark configuration ($\varepsilon = 2\,\text{mrad}$, $\gamma = 3$, $\delta = 1$, clean events, three repetitions per point). (a) The measured extremes of the spectrum of A against the sufficient condition: the Gershgorin floor $\min_i(A_{ii} - R_i) = s - d_{\max}$ (grey) reaches zero once some segment first acquires four compatible neighbours, near $T = 400$, while the true $\lambda_{\min}$ (green) stays between 0.8 and 2.4 all the way to $T = 1000$ and four million segments. The guarantee lapses; the matrix stays positive definite anyway, because it is overwhelmingly sparse. (b) The condition number that follows. (c) The mechanism: the largest off-diagonal row sum saturates at $4 = s$, which is what makes the bound vacuous, while the mean row sum falls towards zero as the isolated population grows.

so eq. (2) is that energy up to a constant independent of $\mathbf{x}$: the same Hessian, the same gradient, the same minimiser, and $\nabla\mathcal{H} = 0$ is eq. (3). What differs between the papers is not the operator but three conventions, and they matter whenever numbers are carried across. First, ref. [1] writes the Ising template $\mathcal{H} = -\frac{1}{2}\mathbf{x}^\top A\mathbf{x} + \mathbf{b}^\top\mathbf{x}$, which for a positive-definite A is $-E(\mathbf{x})$ and hence concave; ref. [3] already uses the convex sign adopted here. Second, their acceptance is a cosine tolerance, $\cos\theta \geq 1 - \varepsilon$, where mine is an angle in radians; the two are related by $\varepsilon_{\cos} = 1 - \cos\varepsilon \simeq \varepsilon^2/2$, so the 10^{-5} of ref. [1] is 4.5 mrad, and the 6.3 mrad of the heavy configuration here is 2.0×10^{-5} in their variable. Third, and most consequential for every closed form below, they take $\gamma = 2$ where this work takes $\gamma = 3$, so their $s = 3$: the isolated attractor sits at $\frac{1}{3}$ rather than $\frac{1}{4}$, a five-hit track at $(\frac{3}{5}, \frac{4}{5})$ rather than $(\frac{4}{11}, \frac{5}{11})$, and a long chain spans $[1, 5]$ rather than $[2, 6]$ — which is why refs. [3, 4] quote a spectrum bounded by $\lambda_{\min} \approx 1$, $\lambda_{\max} \approx 5$ and a filter notch at $\lambda = \gamma + \delta = 3$. Every level in this work is stated at $s = 4$.

The acceptance ε deserves its own derivation, because it is the single most consequential experimental knob and because the formula that sets it is not a convention: it is a quantile of the kink distribution the detector actually produces.

Take three consecutive hits of one true track, on planes $k-1$, k, $k+1$, and the kink between the two segments they form (figure 2a). Two things disturb it. At the middle plane the particle is deflected by multiple scattering, a Gaussian angle of width σ_{scatt} *in each transverse projection*, which adds straight onto the kink. And each of the three hits is recorded with an independent Gaussian position error σ_{res}; in one projection the measured kink is the second difference

$$\theta_x = \frac{x_3 - x_2}{\Delta z} - \frac{x_2 - x_1}{\Delta z} = \frac{x_1 - 2x_2 + x_3}{\Delta z},$$

whose $(1, -2, 1)$ weights give it variance $(1^2 + 2^2 + 1^2)\,\sigma_{\text{res}}^2/\Delta z^2 = 6\,\sigma_{\text{res}}^2/\Delta z^2$: the factor 6 is the stencil, not a fudge. The two sources are independent, so each projection of the kink is Gaussian of width

$$\sigma_p^2 = \sigma_{\text{scatt}}^2 + 6\,\arctan^2(\sigma_{\text{res}}/\Delta z).$$

A track lives in three dimensions, so what the angular cut acts on is the magnitude $\theta = \sqrt{\theta_x^2 + \theta_y^2}$

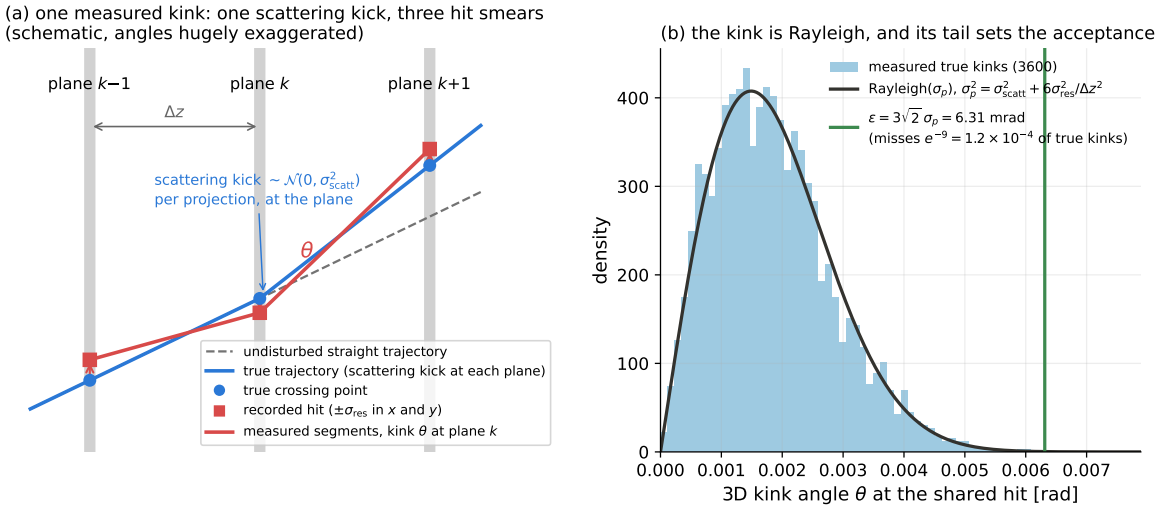


Figure 2: Where the angular acceptance comes from. (a) The geometry of one measured kink: the particle takes a scattering kick in each projection at the shared plane, and all three hits are recorded with an independent position error, so the measured kink is the second difference of three smeared coordinates (schematic; the true angles are a few milliradians and are drawn hugely exaggerated). (b) The resulting three-dimensional kink is Rayleigh-distributed, not Gaussian: the histogram is 3600 true kinks measured on six generated events at the heavy configuration ($\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 20\,\mu$m, $T = 200$), the curve is Rayleigh(σ_p) with σ_p from the closed form and no fit, and the vertical is the acceptance eq. (6) sets, which by construction discards $e^{-9} = 1.2 \times 10^{-4}$ of true continuations.

of two independent Gaussian projections. That magnitude is *Rayleigh*-distributed with scale σ_p, not Gaussian, and it is the Rayleigh tail, $P(\theta > \varepsilon) = \exp(-\varepsilon^2/2\sigma_p^2)$, that decides how many true continuations a cut at ε discards. Figure 2b measures the kink distribution on generated events and finds the closed form holds: at the heavy configuration the measured mean kink is 1.84 mrad against the Rayleigh prediction $\sigma_p\sqrt{\pi/2} = 1.86$ mrad.

Admitting kinks out to κ standard deviations of a projection means $\varepsilon = \kappa\sqrt{2}\,\sigma_p$, which written out is

$$\varepsilon = \sqrt{2\,(\kappa\sigma_{\text{scatt}})^2 + 12\,\arctan^2(\kappa\sigma_{\text{res}}/\Delta z) + 2\,\theta_{\text{min}}^2}\,, \tag{6}$$

with $\theta_{\text{min}} = 15\,\mu$rad a floor that keeps the acceptance finite in the noise-free limit. The coefficients are the per-projection ones, 1 for scattering and 6 for resolution, doubled by the two projections. This work uses $\kappa = 3$ throughout, and the reason to write the cut this way is that it fixes the price: the fraction of true continuations lost is $\exp(-\kappa^2) = e^{-9} = 1.2 \times 10^{-4}$ *whatever* the noise, because ε moves with σ_p. Read backwards, $\varepsilon = \sigma_p\sqrt{2\ln(1/p)}$ sets the acceptance from whatever miss rate p one is willing to pay.

The three configurations give $\varepsilon = 0.42$, 3.18 and 6.31 mrad; the heavy value is fifteen times the clean one, and section 6 shows what that width admits. One configuration in this work does *not* take its acceptance from eq. (6): the benchmark campaign of section 5 holds $\varepsilon = 2$ mrad fixed on clean events, which is 4.7 times the formula value there, $20\sigma_p$ rather than $3\sqrt{2}\,\sigma_p$. Measured on a stored clean event at $T = 200$, the two cuts keep the same 600 true continuations — the formula loses none of them — while the wide one admits 71 false compatibility edges against 5, and at $T = 400$, 661 against 37. The benchmark is therefore a deliberate stress test of the false population, not a physics-derived operating point, and it is labelled as fixed wherever it is used.

2.3 Exact activation levels: the atlas

Because A has constant diagonal, its action decomposes over the connected components of the compatibility graph, and small components can be solved by hand. This is worth doing once in full, because these little closed forms are the vocabulary of every result in this work.

An *isolated* segment (no compatible neighbour) has the scalar equation $(\gamma + \delta)x = \delta$, so

$$x_{\text{iso}} = \frac{\delta}{\gamma + \delta} = \frac{1}{4} = 0.250. \tag{7}$$

This is the false *attractor*: at any multiplicity, the overwhelming majority of the $4T(T-1)$ false segments are isolated and sit at exactly 0.250. A true track, by contrast, is a path P_4 of four segments (five hits), whose matrix is $sI - P_4$ with P_4 the path adjacency. Its solution can be written down exactly:

$$x_{\text{outer}} = \frac{4}{11} \approx 0.364, \qquad x_{\text{inner}} = \frac{5}{11} \approx 0.455, \tag{8}$$

where the outer levels belong to the two end segments (one continuation neighbour each) and the inner levels to the two middle segments (two neighbours each).

These numbers, and the whole family behind them, come from one small calculation that is worth doing explicitly. A chain of m segments has the matrix $sI - P_m$ with P_m the path adjacency, so the j-th row of $(sI - P_m)\,\mathbf{x} = \delta\mathbf{1}$ reads

$$-x_{j-1} + s\,x_j - x_{j+1} = \delta, \qquad j = 1, \ldots, m,$$

a three-term linear recurrence, provided we adopt the convention $x_0 = x_{m+1} = 0$: an end segment simply has no neighbour on one side, which is the same as coupling it to a phantom segment clamped at zero. A recurrence of this kind is solved by any particular solution plus the general homogeneous one. The natural particular solution is the constant

$$\bar{x} = \frac{\delta}{s-2} = \frac{1}{2} \qquad (s = 4,\ \delta = 1),$$

the activation of a segment deep inside an infinitely long chain, where both neighbours carry the same value and the row equation collapses to $(s-2)\,\bar{x} = \delta$. The homogeneous solutions of $-x_{j-1} + s\,x_j - x_{j+1} = 0$ are the geometric sequences $r^{\,j}$ with

$$r + \frac{1}{r} = s \quad \Longrightarrow \quad r = \frac{s - \sqrt{s^2 - 4}}{2} = 2 - \sqrt{3} \approx 0.268,$$

a decaying root and its growing mirror $1/r$. Writing the general solution as $x_j = \bar{x} + A\,r^{\,j} + B\,r^{\,m+1-j}$ (one geometric correction anchored at each end) and imposing the two clamped-end conditions $x_0 = x_{m+1} = 0$ forces, by the chain's reflection symmetry, $A = B = -1/\big(2\,(1 + r^{m+1})\big)$, so the entire activation profile has the closed form

$$x_j = \frac{1}{2} - \frac{r^{\,j} + r^{\,m+1-j}}{2\,(1 + r^{m+1})}.$$

The picture behind the algebra: each end of the chain is a cold boundary that drags the activation below the bulk value $\frac{1}{2}$ by a deficit decaying geometrically into the interior, with penetration depth $1/\ln(1/r) \approx 0.76$ segments. Setting $j = 1$ (or, identically, $j = m$) gives the outer activation

$$x_{\text{outer}}(m) = \frac{1}{2} - \frac{r + r^m}{2\,(1 + r^{m+1})}, \qquad r = 2 - \sqrt{3}, \tag{9}$$

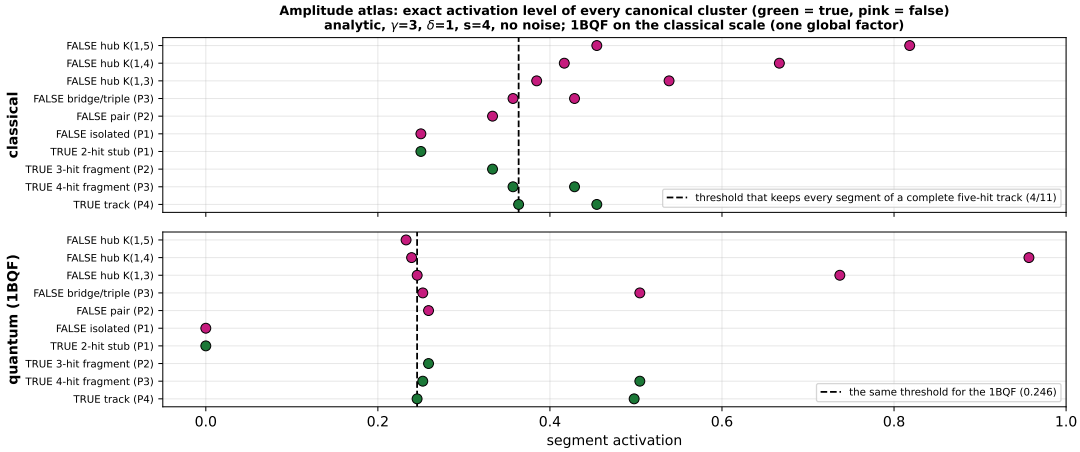


Figure 3: The amplitude atlas: exact activation level of every canonical cluster type (rows), classical (top) and after the 1BQF (bottom). This figure is analytic: it is the exact solution of $A\mathbf{x} = \delta\mathbf{1}$ on each cluster in isolation at $\gamma = 3$, $\delta = 1$, $s = 4$, with no events, no noise and no acceptance; the filtered amplitudes are the response $f(A)\mathbf{b}$ of eq. (12) placed on the classical scale by one global factor, so the two panels are directly comparable. The dashed line is the threshold that would keep every segment of a *complete* five-hit track, 4/11 classically and 0.246 after the filter: on clean events, where every true segment belongs to such a track, it is the efficiency-first threshold of the analytic picture. Two things are then visible that the rest of this work spends its time on. The true fragments of broken tracks fall *below* that line, on top of the false levels. And every cluster pair with the same graph carries exactly the same amplitudes in both panels: a true two-segment fragment sits on the false pair, a true three-segment fragment on the false bridge, and the one-segment stub sits on the isolated false segment at $\lambda = s$, so the filter's single zero annihilates the two together — it removes every isolated fake, and every one-segment true fragment with them. Green: true; pink: false. Cluster names and their graphs are defined in section 2.5; a stub is a one-segment true fragment (P_1) and a hub is a star ($K_{1,m}$).

and $j = 2$ at $m = 4$ returns the inner 5/11 of eq. (8). Equation (9) climbs the ladder $\{0.250,\ \frac{1}{3},\ \frac{5}{14},\ \frac{4}{11}, \dots\} \nearrow (1-r)/2 = (\sqrt{3}-1)/2 \approx 0.366$ for $m = 1, 2, 3, 4, \dots$: a short chain sits low because the two boundary deficits overlap, and the ladder saturates once the ends are further apart than the penetration depth. It is worth working two rungs by hand. A two-hit fragment (m=1) sits at exactly the isolated-false attractor 0.250, and a three-hit fragment (m=2) sits at 1/3, only 0.083 above it. Fragments are barely distinguishable from noise by amplitude alone; this innocent-looking fact returns in section 9 as the ceiling on segment efficiency.

The decision rule is a threshold: segment i is *active* iff $x_i > \tau$. Every threshold choice trades efficiency against false rate, and this work reads every solver, the classical inversion included, at one working point, the *efficiency-first* one: the threshold is the lowest value that admits the true solutions, placed just below the 1% quantile of that solver's own true-segment amplitudes so that 99% of true segments survive by construction, and the false rate at that threshold is the price quoted. An earlier convention, inherited from ref. [3], fixed an absolute threshold $\tau = \delta/(\gamma+\delta) + 0.10$ (0.35 at $\gamma = 3$) as a margin above the false attractor. It is not used here: a filter reshapes the amplitude atlas, and that fixed cut lands inside the one-bit filter's reshaped true band (section 3), discarding true segments and understating the false rate at the same time. Section 3 shows the full trade-off explicitly in figure 6.

2.4 Metric definitions

Every number in this work is one of the following, and I will never use these words in any other sense:

- **segment efficiency** $\varepsilon_{\text{seg}} = n_{\text{true active}}/n_{\text{true}}$: the fraction of true segments that survive selection;
- **segment false rate** $f_{\text{seg}} = n_{\text{false active}}/n_{\text{active}}$: the fraction of the *selected* sample that is fake (note the denominator: this is an impurity, not a false-positive rate over all fakes);
- **working point**: the *efficiency-first working point*, the lowest threshold that admits the true solutions: τ placed just below the 1% quantile of the solver's own true-segment amplitudes, so $\varepsilon_{\text{seg}} \geq 0.99$ by construction. Whenever this work quotes a false rate without qualification, this 99% point is meant, for every solver. Sections 8–9 use the same rule at other target efficiencies, there called *matched-efficiency*: τ just below the k-th largest true amplitude with $k = \lceil \text{eff} \cdot n_{\text{true}} \rceil$, which lets solvers be compared at identical efficiency without quantile-grid artefacts. Because the threshold is a quantile of the solver's own amplitudes, these metrics do not depend on the amplitude scale.
- **AUC**, the threshold-free separation metric: sweeping τ from high to low traces out a curve of efficiency against false rate for each solver, and the *area under that curve* summarises how well the solver's amplitude ordering separates true from false segments independently of any particular threshold choice. AUC = 1 means some threshold separates them perfectly; AUC = 0.5 means the ordering is no better than chance. It is used when a single number for "separation quality" is needed, notably in section 7.

Quantum solutions are unit-norm vectors; before thresholding they are rescaled to the classical amplitude scale on the classically-active support. Rescaling by the full norm would be multiplicity-dependent, since the false bulk owns most of the norm. All thresholds act on $|x_i|$.

2.5 The spectrum

Because $A = sI - C$, its spectrum is a rigid reflection of the compatibility graph's: $\lambda(A) = s - \mu(C)$, where $\mu(C)$ denotes an eigenvalue of C, the ε-windowed 0/1 compatibility adjacency of section 2.2 ($C_{ij} = 1$ exactly when segments i and j continue one another head-to-tail within the angular acceptance ε, and 0 otherwise). Every spectral statement in this work is therefore a statement about the eigenvalues of small 0/1 graphs. Each cluster type gets a name and, once and for all, the graph that name stands for; later sections carry both, so that no name has to be remembered on its own. Three families matter:

1. **Isolated segment**, the graph P_1 (one vertex, no edge): a segment with no compatible continuation. $\lambda = s$ exactly. This single line carries $\mathcal{O}(n_{\text{seg}})$ modes: at $T = 200$ under heavy noise, 155,371 of 157,609 segments (rep 0).
2. **Chains**, the path graphs P_m (m vertices in a line): m segments each continuing the next. $\lambda_k = s - 2\cos(k\pi/(m{+}1))$, $k = 1, \ldots, m$. Short chains have their own names, which are the ones the figures use: a *pair* is P_2, two segments meeting at a shared hit; a *triple* is P_3. Note that a triple arises two ways that the matrix cannot tell apart — three segments in a row, or one segment continued by two others — because both are the graph P_3; this is the first appearance of the degeneracy that section 5.5 turns into a theorem. A true track ($m = 4$) therefore lives on exactly four lines,

$$\lambda \in \{s - 2\cos(k\pi/5)\}_{k=1}^{4} = \{2.382,\ 3.382,\ 4.618,\ 5.618\} \quad (s = 4), \tag{10}$$

   symmetric about s and split by the golden-ratio cosines[2].

[2] The values $2\cos(\pi/5) = \varphi = 1.618\ldots$ and $2\cos(2\pi/5) = 1/\varphi$: the five-hit track is secretly a pentagon problem.

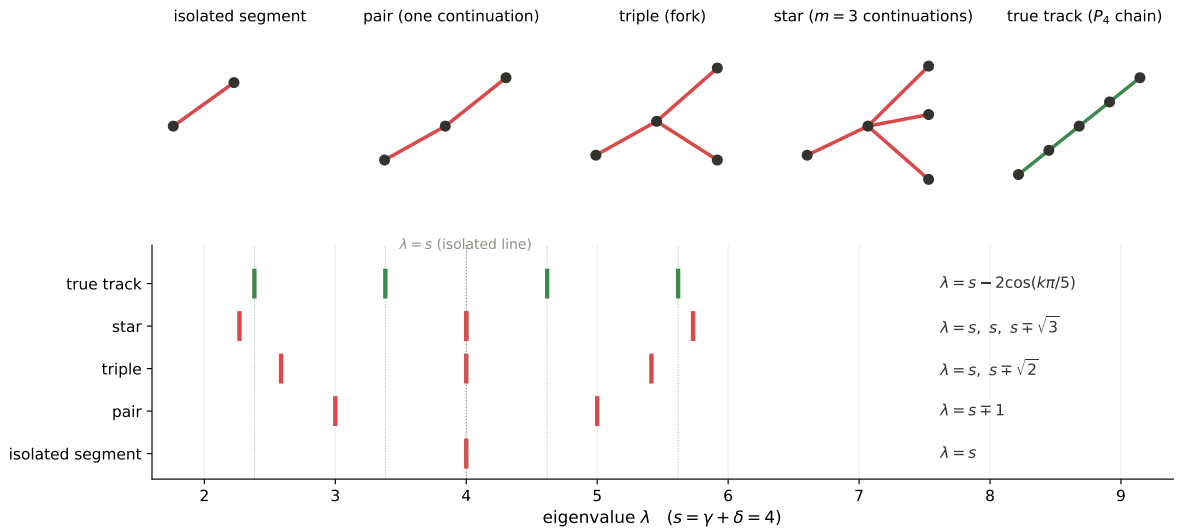


Figure 4: The motif dictionary. Top: the canonical segment clusters drawn in hit space (dots are hits, lines are segments; green true, red false). Bottom: the exact eigenvalue lines each motif contributes, from the closed forms of section 2.5, at $s = \gamma + \delta = 4$. Every false family sits close to, but never on, the four true-track lines (green dotted), except for the spectral twins of section 5.5.

3. **Stars**, the complete bipartite graphs $K_{1,m}$ (a centre joined to m leaves and the leaves joined to nothing): one segment continued by m different others, which do not continue one another. $\lambda = s \pm \sqrt{m}$ plus $m-1$ copies of s.

Figure 4 draws each of these motifs in hit space and places its exact eigenvalue lines on a common λ-axis: this is the dictionary between what a cluster of segments *looks like* in the detector and where its solution weight *piles up* in the spectrum. The pair motif at $\lambda = s \pm 1$ and the triple at $\{s,\ s \pm \sqrt{2}\}$ do most of the false-population work under noise, and their lines interleave the true lines of eq. (10) without touching them. That gap, false lines close to but not on the true lines, is what this work exploits. When noise is switched on in section 6, the *measured* spectral pile-up of a heavy-noise event (figure 12) lands exactly on the lines catalogued here. Motifs whose spectrum coincides with a true motif's are the exception, and they return in sections 5.5 and 9 as the limiting factor.

3 The one-bit quantum filter, read as a spectral filter

3.1 The circuit and its response function

The 1BQF of ref. [3] is a one-ancilla interference experiment. Prepare the uniform superposition $|u\rangle$ over the (zero-padded) segment register, put the ancilla in $(|0\rangle + |1\rangle)/\sqrt{2}$, apply e^{iAt} to the register controlled on the ancilla, close the interferometer with a second Hadamard, and keep the runs in which the ancilla reports the interference branch. The kept state is

$$|\psi\rangle \;\propto\; \tfrac{1}{2}(I + e^{iAt})|u\rangle \;=\; \sum_i e^{i\lambda_i t/2} \cos\!\Big(\frac{\lambda_i t}{2}\Big)\, \langle v_i|u\rangle\, |v_i\rangle, \tag{11}$$

where $\{\lambda_i, |v_i\rangle\}$ are the eigenpairs of A. Reading out $|\langle j|\psi\rangle|$ segment by segment gives amplitudes x_j^Q, and the whole device is summarised by a single real response function applied to the spectrum:

$$f(\lambda) = \cos\!\Big(\frac{\lambda t}{2}\Big), \qquad t = \frac{\pi}{s}. \tag{12}$$

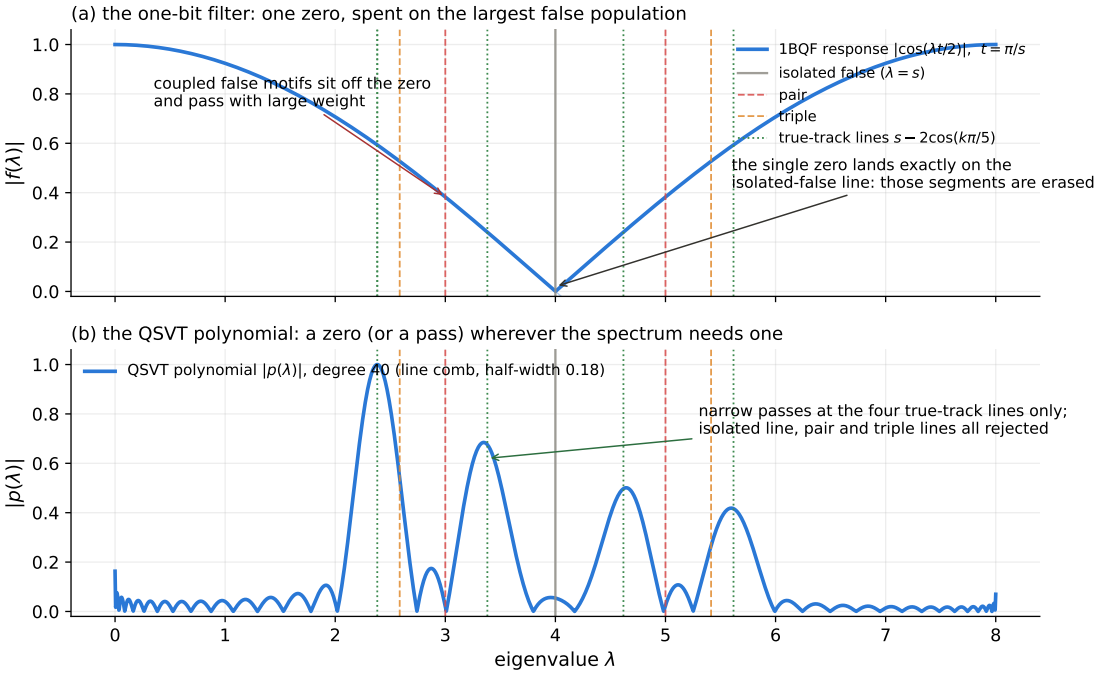


Figure 5: The two response functions against the populated spectrum ($s = 4$). (a) The 1BQF cosine: one zero, placed on the isolated-false line; the coupled false motifs (pair, triple) sit on the slopes and pass. (b) The degree-40 line comb: passes at the four true-track lines only, with the isolated, pair and triple lines all rejected.

The choice $t = \pi/s$ places the zero of the cosine exactly at $\lambda = s$, the isolated-segment line. Every isolated false segment, all $\mathcal{O}(n_{\text{seg}})$ of them, is annihilated exactly: not suppressed, killed. This is the entire content of the 1BQF, and the form in which we will need it later is a simple accounting rule: each application of the filter buys exactly one zero of the response, and the 1BQF spends its one zero on the largest false population in the problem. The padding modes introduced to reach a power-of-two dimension are diagonal-only, sit at $\lambda = s$ too, and are removed by the same zero, a small but pleasing piece of engineering hygiene.

Figure 5 makes the geometry of this choice, and of everything that follows, visible. The top panel is the 1BQF response of eq. (12) drawn over the populated spectrum: its single zero sits exactly on the isolated-false line, and every other population, the pair and triple motif lines included, lands on the slopes of the cosine and passes with large weight. The filter attacks the isolated segments and nothing else; whatever false weight is *coupled* sails through. The bottom panel shows the QSVT alternative developed from section 4 onward: a degree-40 polynomial with narrow passes at the four true-track lines and effective zeros everywhere else, including the isolated, pair and triple lines simultaneously. The one-sentence summary of this work is the difference between those two panels: a degree-one response owns one zero, a degree-d response owns d, and the noisy-event false spectrum demands more than one.

3.2 What the notch cannot do

Away from its zero the cosine is not flat, and this has two consequences that define the 1BQF's operating envelope.

The first is that true amplitudes are reshaped. A true track's four modes (eq. (10)) sit symmetrically about s, where $|f(\lambda)|$ is small; the filtered chain no longer reproduces the classical $\{4/11, 5/11\}$ profile. The effect has a closed form: the outer (endpoint) segments of a clean track drop from 0.364 to 0.246 while the inner pair rises slightly, from 0.455 to 0.498 (figure 3, on

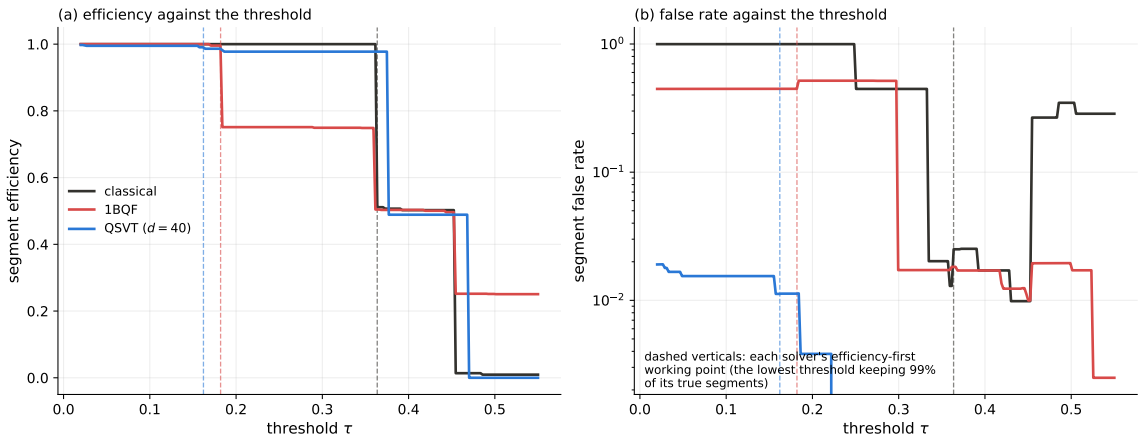


Figure 6: The threshold trade-off, on the stored $T = 400$ clean event of the canonical campaign: (a) segment efficiency and (b) segment false rate as the threshold τ sweeps, for the classical solver, the 1BQF and the QSVT (degree 40, base A; quantum amplitudes on the classical signal scale). The dashed verticals are each solver's efficiency-first working point (the lowest threshold keeping 99% of its true segments). The 1BQF's efficiency steps down at its reshaped endpoint band near $\tau \approx 0.28$, so its working point sits below that band, while the QSVT holds both high efficiency and a $\sim 10^{-2}$ false rate over the whole usable range.

the common scale defined there). This is why the fixed cut of the earlier convention (0.35) was wrong for the filter: it sat above the endpoint band and reported a segment efficiency plateau of ≈ 0.73–0.75, finding the middles of tracks and losing the ends. The endpoint loss was a property of that cut, not of the physics; the endpoint amplitudes remain perfectly well ordered above the false ones, so the efficiency-first threshold, which sits below the endpoint band, recovers them all.

Figure 6 shows the whole trade explicitly, on a stored $T = 400$ clean event, by sweeping the threshold for all three solvers of this work. The efficiency panel shows the plateau structure just described: the 1BQF efficiency steps down at the endpoint band around $\tau \approx 0.28$, so any cut above it reads 75%, not because the filter lost the physics but because the cut sits above a reshaped true band. The false-rate panel shows what moving the threshold costs. Every solver is read at its own *efficiency-first working point* (dashed verticals): the threshold just below the 1% quantile of that solver's true amplitudes, so that 99% of true segments survive by construction, with the false rate quoted as the price. For the 1BQF that threshold is small, because the filter has already sent the isolated false segments to zero. On clean fixed-acceptance events at $\gamma = 3$ the price is steep at high multiplicity: the 1BQF false rate climbs to several tens of percent by $T = 400$ and beyond, and section 5 quantifies this against the comb.

The second consequence is that coupled false segments pass. A false segment that accidentally couples to the graph moves off the $\lambda = s$ line, out of the zero, and into the passband. The notch has no second zero to spend on it. As density or acceptance width grows, the coupled-false population grows, and the 1BQF's false rate grows with it. The 1BQF is, precisely, the statement that unconnected fakes are exactly removable; everything in this work beyond section 4 is about the connected ones.

On success probability: the kept-branch probability scales as $P_{\text{anc}} \approx 0.25/T$ on this problem, since the uniform input spreads over $4T^2$ segments of which order T carry surviving amplitude, so shot budgets grow linearly in T and amplitude amplification buys the usual quadratic discount.[3]

[3]The 0.25 constant is the analytic estimate for the uniform input; the comb's analogous constant, ≈ 0.44, is directly measured (section 5.4).

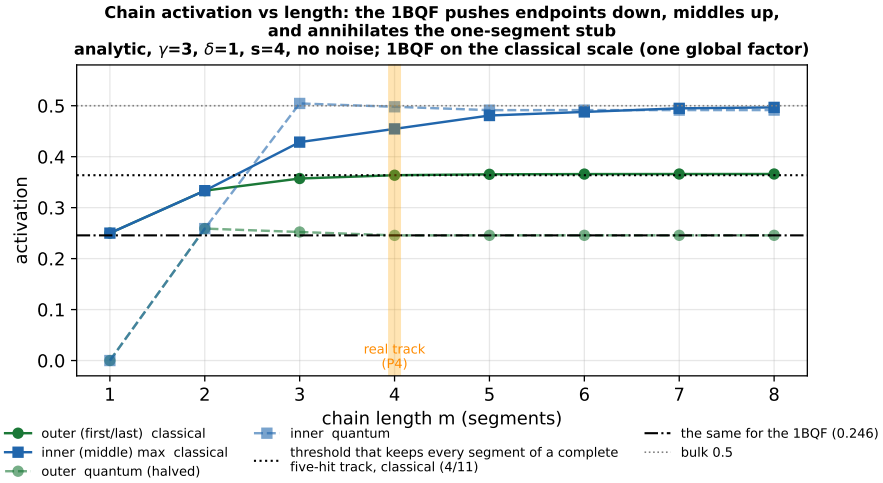


Figure 7: The chain ladder, on the same analytic footing as figure 3 ($\gamma = 3$, $\delta = 1$, $s = 4$, no noise, the filtered amplitudes on the classical scale by the same global factor). Classical outer activation climbs eq. (9) toward 0.366; the 1BQF pushes the outer level of a five-hit track down to 0.246 and the middles up to 0.498, and annihilates the one-segment stub outright, since a lone segment sits exactly on the notch. The marked levels are what a threshold has to clear to keep every segment of a complete track, and the reason the earlier fixed cut at 0.35 lost the filtered chain's endpoints.

These costs carry over essentially unchanged to the polynomial filters below, which is why I defer resource accounting to section 5.4.

4 From one bit to any polynomial: the QSVT construction

The promotion from eq. (12) to an arbitrary polynomial response rests on three standard ingredients: block encoding, qubitization, and a linear combination of unitaries (LCU). I assemble them here in the concrete, real-arithmetic form used by the simulations. Readers who trust the QSVT literature [7–10] can skim to the resource summary in section 4.3; the details are included because this work's cost claims (qubit counts, walk calls, success probabilities) are read directly off this construction.

4.1 Block encoding and the qubitization walk

Rescale the Hamiltonian affinely so its spectrum fits in $[-1, 1]$: $X = (A - c_0 I)/c_1$ with c_0 the spectrum centre and c_1 slightly more than its half-width[4]. A block encoding is a unitary that hides X in its top-left corner:

$$U = \begin{pmatrix} X & \sqrt{I - X^2} \\ \sqrt{I - X^2} & -X \end{pmatrix}, \tag{13}$$

acting on the segment register plus one *dilation* qubit. The picture I find most useful is a rotation in each eigenplane: for every eigenvalue x of X there is a two-dimensional invariant subspace in which U rotates by the angle $\arccos x$.

[4]The implementation pads the user-supplied spectral bounds by a few percent; an early campaign bug taught us that the polynomial's fit domain must exceed the padded bounds or the constructor rightly refuses.

The word *walk* deserves an explanation, because nothing in this work has moved yet and the term is imported jargon. A walk operator is simply a fixed unitary that one applies over and over, taking one "step" per application, in analogy with a classical random walk where one transition matrix is applied repeatedly. The useful property is what repetition does. Because U rotates each eigenplane by $\arccos x$, applying the walk k times rotates it by $k \arccos x$, and the component that ends where it started is $\cos(k \arccos x)$, which is precisely the k-th Chebyshev polynomial $T_k(x)$. One application of the walk knows only x; k applications know $T_k(x)$. Repetition of a single fixed circuit therefore *generates a polynomial family*, one degree per step, and that is the entire mechanism by which a quantum computer applies polynomials of a matrix without ever diagonalising it. Formally, the *qubitization walk* $W = (Z \otimes I)\, U$ (the extra sign Z on the dilation qubit turns the reflections of U into clean rotations) has the defining property

$$\langle 0|\, W^k\, |0\rangle_{\text{dil}} = T_k(X), \tag{14}$$

where T_k is the k-th Chebyshev polynomial and the bra-ket projects the dilation qubit onto $|0\rangle$. Equation (14) is exact, with no Trotterization and no error terms, and every operator in it is real, so the entire simulation can run in real arithmetic on orthogonal matrices. Walking k steps and looking at the undeflected component applies the k-th Chebyshev polynomial of the Hamiltonian; that is the whole theorem.

4.2 Summing the series: LCU

A degree-d response $p(X) = \sum_{k=0}^{d} c_k T_k(X)$ is then a weighted sum of walk powers. Introduce an index register of $m = \lceil \log_2(d{+}1) \rceil$ qubits, prepare $|\text{prep}\rangle \propto \sum_k \sqrt{|c_k|}\, |k\rangle$, apply W^k controlled on the index (in practice: controlled powers W^{2^j}), undo the preparation, and post-select the index and dilation registers on zero. The surviving register state is $\propto p(X)|u\rangle$ with success probability

$$P_{\text{anc}} = \frac{\left\| p(X)\, |u\rangle \right\|^2}{\left(\sum_k |c_k| \right)^2}, \tag{15}$$

the usual LCU price: the one-norm of the coefficient vector. The 1BQF is recovered, exactly, as the $d = 1$ row of this table (cos is T_1 of the appropriately rescaled operator), which is the formal content of the claim that this work continues rather than replaces its predecessors.

Simulation-side, eq. (14) gives a matrix-free evaluator: $p(A)\mathbf{b}$ by the Chebyshev three-term recursion on sparse matrix-vector products, $\mathcal{O}(d \cdot \text{nnz}(A))$ flops. In a persisted benchmark of the production configuration the $d = 40$ comb solve runs in 1.2–1.7 s at $T = 1000$ ($n = 4 \times 10^6$), and the evaluator agrees with the full qiskit circuit and with a gate-streaming evaluation to better than 10^{-14} at circuit-affordable size (three-way maximum discrepancy 8×10^{-15}). All large-T results below use this exact evaluator; nothing in this work relies on an approximate simulation of the quantum readout.

4.3 Resource summary

For a degree-d filter on an event of T tracks:

- **Width**: $n_{\text{sys}} = \lceil \log_2 4T^2 \rceil$ system qubits $+1$ dilation $+ m = \lceil \log_2(d{+}1) \rceil$ index qubits. With the production degree $d = 40$ this is 21 qubits at $T = 50$ rising to 29 at $T = 1000$. Width is cheap, and grows only logarithmically.
- **Depth**: d controlled walk calls per coherent repetition, times $\mathcal{O}(1/\sqrt{P_{\text{anc}}})$ amplitude-amplification rounds.
- **Success**: $P_{\text{anc}} \propto 1/T$ (measured constant ≈ 0.44 for the comb of section 5), so total walk calls scale as $d\sqrt{T}$.

The quantum content of this work is this resource profile; the response functions themselves are classically simulable by the same recursion, and I return to what that does and does not concede in section 11.8.

5 Fitting the polynomial to the clean events: the line comb

5.1 The design insight: the true spectrum is discrete

Here is the observation that turns QSVT from a mathematical possibility into a design tool. On this detector geometry every true track has exactly five hits, hence exactly four segments, hence, by eq. (10), exactly the same four eigenvalues. The true spectrum of the entire event, however many thousand tracks it contains, is four discrete lines. We are not trying to pass a region of the spectrum; we are trying to pass four known radio stations, and everything between them is static.

My first design did not understand this. A band-limited approximate inverse ($\approx 1/\lambda$ across a contiguous band containing the true lines, zero outside) works beautifully at low multiplicity and then fails exactly where it matters: as T grows, accidental tangles of false segments produce eigenvalues throughout the band, and a band design passes them all. Its false rate reached tens of percent by $T = 400$, worse than classical.[5] The production design is therefore a *line comb*: narrow Gaussian passes of half-width h_w, weighted $\propto 1/\lambda$, centred at the four lines $s - 2\cos(k\pi/5)$ only, fitted as a bounded Chebyshev series of degree d. The defaults, fixed once by a two-dimensional (d, h_w) scan and never tuned per event, are $d = 40$, $h_w = 0.18$. This degree-40 line comb on the base Hamiltonian A is *the QSVT* of this work, the comb, so called because it combs the false tracks out of the spectrum: wherever the text says "the QSVT" or "the comb" without qualification, this polynomial on A is meant. Every other QSVT result is labelled by its polynomial degree and by the Hamiltonian it is set up on, and where a polynomial of the same degree has its coefficients refit to the events (section 9) the text says so.

Why does a width of 0.18 work? The nearest systematically-populated false lines, the P_3 bridge modes at $\{s - \sqrt{2},\ s,\ s + \sqrt{2}\}$, sit $\gtrsim 0.2$ away from the true lines, so the comb threads the gap; and the resolution law of a degree-d Chebyshev series, features no narrower than $\approx 2W/d$ on a domain of half-width W, makes $d = 40$ sufficient for $h_w = 0.18$ on the clean-event domain. Degree, width, and line spacing are one budget, and this is the budget the noisy events of section 9 will re-open.

5.2 Clean-event results

On the fixed-acceptance ($\varepsilon = 2\,\text{mrad}$), $\gamma = 3$ study, the canonical benchmark of the programme, with events and solutions shared segment-for-segment across all three solvers through a common data pipeline, every solver read at the efficiency-first working point of section 2.4 (99% of its true segments kept), the picture is:

- **Classical**: efficiency 100% throughout (the true amplitudes are tightly clustered, so the threshold that keeps 99% keeps them all); false rate grows from ≈ 0 at low T to 0.65% at $T = 200$, 1.49% at $T = 400$, 4.9% at $T = 700$ and 10.0% at $T = 1000$ as the $\mathcal{O}(T^2)$ false pairing feeds the active set.
- **1BQF**: the notch recovers essentially every true segment (efficiency 99.5–100% at every T), and the false rate blows up to pay for it: 5.1% at $T = 100$, 43.9% at $T = 400$, 82.6% at $T = 1000$, far above the classical solver, because the single zero removes only the isolated-false bulk while every coupled fake sits off the notch and survives.

[5] A useful failure: it taught us that the exploitable structure is the discreteness of the geometry-pinned true spectrum, not its location.

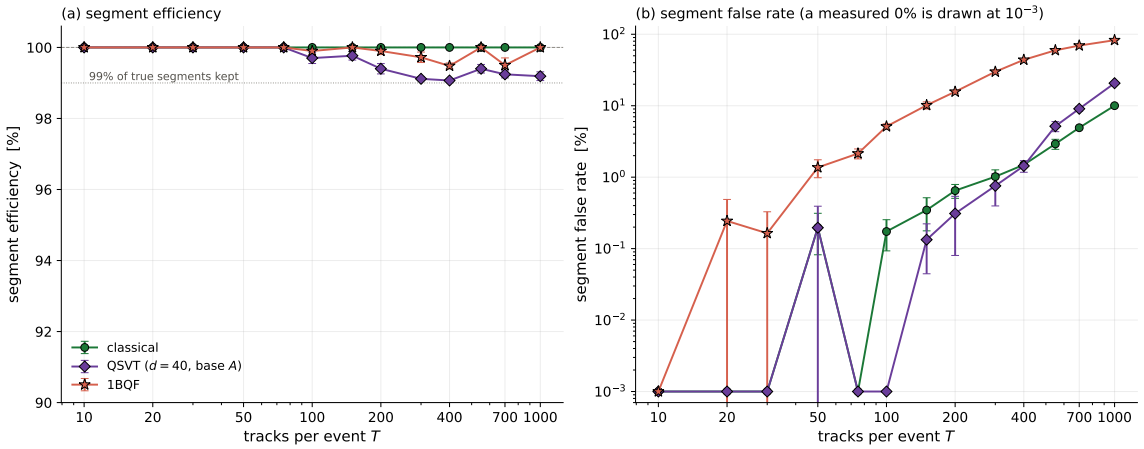


Figure 8: The clean-toy headline ($\gamma = 3$, fixed $\varepsilon = 2\,\text{mrad}$, no hit drop): (a) segment efficiency and (b) segment false rate against multiplicity, for the classical solver, the QSVT (degree 40, base A) and the 1BQF, every solver at its efficiency-first working point (99% of its true segments kept; the efficiency axis shows the convention holding). Error bars are standard errors over event repetitions (classical 5–20, QSVT 5–10, 1BQF 3–10 per T); a measured 0% false rate is drawn at 10^{-3}.

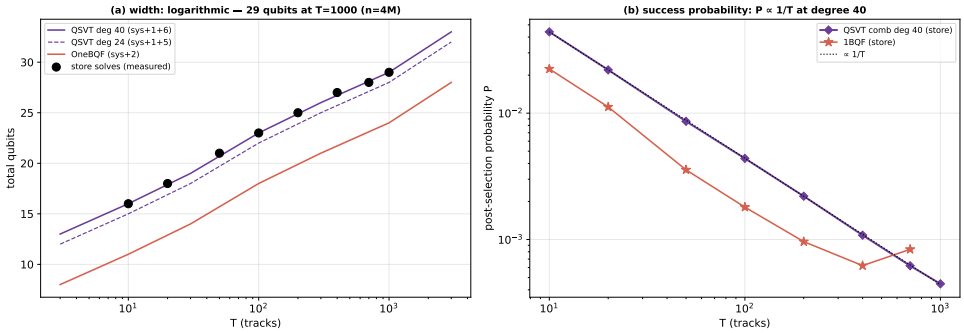
- **QSVT**: efficiency 99.1–100% with a false rate of 0.31% at $T = 200$ against the classical 0.65% (zero to a few false actives per event where the classical solver admits tens), a tie at $T = 400$ (1.44% against 1.49%), and then behind: 9.1% at $T = 700$ and 20.7% at $T = 1000$ against the classical 4.9% and 10.0%.

The comb wins clearly through $T = 300$ and ties at $T = 400$; beyond that, at this acceptance, it loses. Two things happen at once. The composition effect dominates everyone: the false-to-true ratio $n_F/n_T \approx T - 1$ converts even excellent per-segment separation into a growing impurity. And the clean-event design itself stops being right: from $T \approx 400$ the 2 mrad events develop tangles of false segments whose eigenvalues enter the comb's passes (section 5.5), so a polynomial pinned to the isolated-chain lines no longer separates. The remedy is not a different solver but the same degree-40 polynomial with its coefficients refit to the events, the subject of section 9 and, on these very events, of figure 24: refit, it reads 2.8% at $T = 1000$ against the classical 10.0%. One qualifier matters enough to state here rather than in a caption: the wall is set by the *acceptance*, not by multiplicity alone. This whole section runs at the wide fixed $\varepsilon = 2\,\text{mrad}$ of the benchmark configuration, whose window admits several times more false candidates than the formula acceptance of eq. (6); at the formula acceptance the same $T = 1000$ multiplicity is solved outright (99.98% efficiency at a false rate below 0.1%, section 10) — a direct measurement of section 2.2's claim that ε is the most consequential experimental knob.

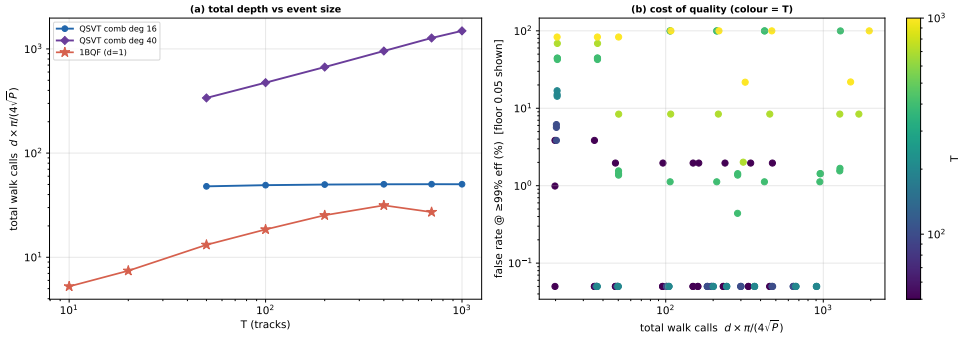
The comb's amplitude picture explains the numbers. Under the comb response the false population collapses to $\lesssim 10^{-2}$ of the true median (the passes reject everything off the lines), while the true band stays tight around its classical profile: the $1/\lambda$ weighting inside the passes preserves the chain shape rather than reshaping it as the single cosine does. Where the 1BQF splits the true population and the classical solver drags a false tail across the threshold, the comb simply leaves a gap.

5.3 Hit inefficiency: the first appearance of fragments

Dropping 1% of hits breaks $\approx 5\%$ of tracks into fragments: chains of one, two, or three segments whose spectra (section 2.5) are *not* on the four P_4 lines. The comb, by design, does not pass them. Per-class efficiency under the production comb is 100% on intact tracks and 0% on every



(a) Register width and success probability.



(b) Total walk calls (depth).

Figure 9: Measured resource scaling of the comb on the shared events: width grows logarithmically (21 qubits at $T = 50$ to 29 at $T = 1000$); success probability falls as $\approx 0.44/T$; amplified walk calls grow as $d\sqrt{T}$, with the half-resolved $d = 16$ comb holding a T-independent ≈ 50-call budget to $T = 700$.

fragment class, which is why its overall efficiency reads 94.0% rather than 97–98% at $T = 400$ under drop. Extending the comb with passes at the fragment lines does not help: it re-admits their false spectral twins wholesale (a P_2 fragment and a false pair are the same matrix, figure 10), so fragments are a spectral floor *on this Hamiltonian*, the first appearance of the twin obstruction that section 9 meets at full strength, and that section 9.4 shows an operator modification can lift. Recovery on the unmodified Hamiltonian must come from information a spectral filter cannot see, namely hit-level constraint information, and that is classical post-processing, outside the scope of this work.[6]

5.4 Resource scaling and the depth ladder

Measured on the shared events: total width 21 qubits at $T = 50$ to 29 at $T = 1000$; success probability $P_{\text{anc}} \approx 0.44/T$; walk calls per amplified repetition $d \cdot \pi/(4\sqrt{P_{\text{anc}}}) \propto d\sqrt{T}$. Two findings sharpen the budget. A half-resolved comb at $d = 16$ matches the $d = 40$ metrics up to $T = 700$ at an approximately T-independent $P_{\text{anc}} \approx 6\%$, about fifty total walk calls, flat in T, because partially-merged passes still cover the four lines while rejecting the bulk. And the degree axis is not monotonic: intermediate degrees ($d \approx 20$–32) can cliff at the efficiency-first working point through Chebyshev fit pathologies, so every production degree must be validated, not interpolated.

[6] An earlier version of this study paired a hit-occupancy recovery gate with the comb alone; it was retired as mission creep (no other solver received the same help, and the tool is orthogonal to the spectral question). Classical post-selection is developed and reported separately in the programme's occupancy study.

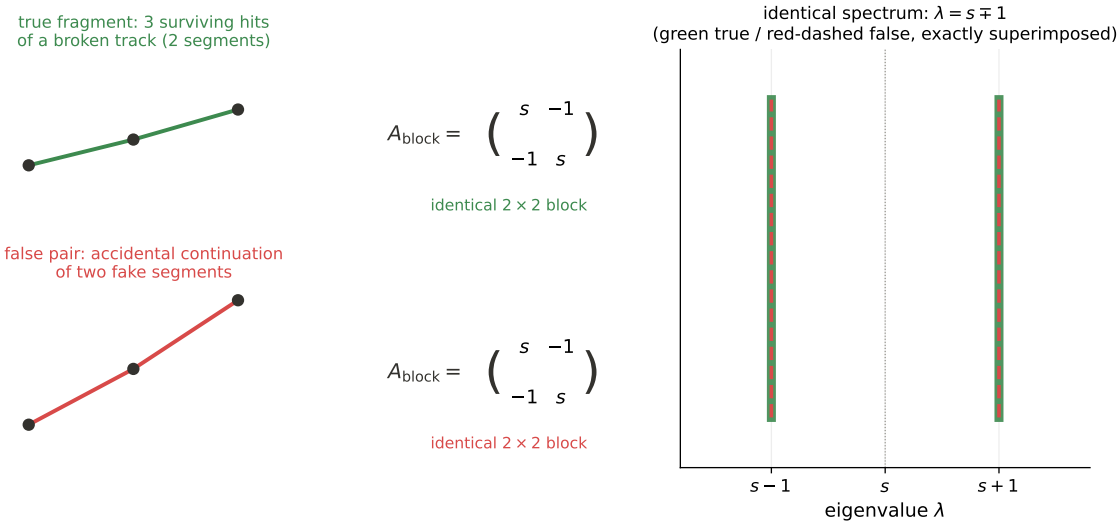


Figure 10: Spectral twins. A true fragment of a broken track (top) and an accidental false pair (bottom) have isomorphic compatibility graphs, hence identical 2×2 sub-blocks of A, hence exactly the same eigenvalues and amplitudes. Every function of A treats them identically; this is the floor theorem, and under noise these twins become a population large enough to cap segment efficiency (section 9).

5.5 Spectral twins and the floor theorem

The most important structural limit on the whole filter family deserves its own name and its own picture. Call two segment clusters *spectral twins* when their compatibility graphs are isomorphic: the same number of segments, connected in the same pattern, regardless of where the segments physically are or whether they belong to a real track. Because A acts block-diagonally over connected components, each cluster contributes a sub-block of A, and twin clusters contribute *the same sub-block up to relabelling* (permutation similarity). The consequence is the *floor theorem*: for every function f, $f(A)\mathbf{b}$ assigns twin clusters identical amplitudes, because permutation similarity commutes with matrix functions and $\mathbf{b}$ is uniform on every cluster. No response, fitted or otherwise, and no classical amplitude scoring, can tell twins apart: whatever passes one passes the other.

Figure 10 draws the pair of twins that will matter most in this work. A true two-segment fragment (three surviving hits of a broken track) and a false pair (two fake segments that happen to continue one another inside the acceptance) are physically completely different objects; but both are two segments sharing one hit, both produce the same 2×2 sub-block of A, and both therefore sit on exactly the eigenvalue lines $s \mp 1$ with exactly the same amplitudes. On clean events twins are rare (an exhaustive scan of the $T \leq 200$ events finds none; 0.6% of true segments by $T = 400$ with the acceptance widened to 4 mrad), but they are the true ceiling of the design, and under noise they return, as a population, in section 9.

Two further structural results complete the clean-toy picture. The *shared-line theorem*: the odd modes of a contaminated five-segment chain coincide exactly with the $s \pm \sqrt{3}$ lines of the three-pronged hub $K_{1,3}$, so no comb can null both populations independently; pass menus trade one against the other with a fixed exchange rate. And the *mirror tax*: a chain-reflection symmetry forces the false extension of a recovered cluster to the recovered-true amplitude level, so recovery passes always buy their efficiency at a false-rate premium. Together these results closed the clean-toy design question. The production comb is optimal up to provable exceptions, and further progress must come from changing the operator or leaving the spectral domain, which is exactly where the rest of this work goes.

6 The heavy-noise events

From here on the operating point is the heavy configuration: $\sigma_{\text{scatt}} = 10^{-4}$, $\sigma_{\text{res}} = 20\,\mu$m, hit drop 1%, at $T = 200$, with the acceptance set by the kink formula to $\varepsilon = 6.3$ mrad. Three reps of every event are used throughout; where a single event is dissected, it is rep 0.

The acceptance width is the story. At $\varepsilon = 2$ mrad a false segment rarely finds an angular partner; at 6.3 mrad the compatibility graph fills with small false structures. Two facts about that graph, both established by direct decomposition of the solved events, control everything downstream:

1. **There is no giant tangle.** The compatibility graph at heavy $T = 200$ splits into 869 nontrivial connected components of at most 17 segments, floating in a sea of 155,371 isolated segments (rep 0). Noise, at this operating point, buys many small motifs rather than one big mess, and small components mean the entire spectrum is computable exactly, component by component, with dense eigensolvers. Every spectral statement in sections 7–9 is exact, not approximate. Figure 11 shows what these components actually look like in the detector: the component-size distribution, and real examples drawn from the event, from a pure-false pair to the largest seventeen-segment tangle. Seen at the micron scale of their internal kinks, the "tangles" are bundles of near-collinear competitors, exactly the structures the widened acceptance admits.

2. **The false actives are motif-borne.** At the classical efficiency-first working point the event has 2238 active segments of which 1454 are false; those false actives live in 707 components that contain, between them, only 142 true segments. Ninety percent of the false-active population sits in essentially pure-false motifs (pairs, triples, short bridges), not entangled with true tracks at all. Spectrally, this population lands exactly where the motif dictionary of figure 4 says it must: figure 12 shows the measured per-mode solution weight of the event against λ, with the false weight piled up on the pair line $s - 1$ and the triple lines $s \pm \sqrt{2}$, interleaving but not touching the four true lines.

The solver scoreboard at this point makes uncomfortable reading. At the efficiency-first working point the classical false rate is 0.65; the 1BQF's is 0.65 as well, since the notch's one zero still annihilates the isolated bulk while every false *active* is by definition coupled, off the line, and untouched. The clean-event comb, designed for a 2 mrad world where false structures barely exist, fails too: the wide acceptance populates false motif lines (figure 12) inside its pass windows. The important statement here is the diagnosis: at heavy noise all three solvers fail for the same reason, a false population that has moved onto spectral lines the designs were never told about, and the cure, developed in section 9, is to measure those lines and refit the response to them. The classical literature also offers two Hamiltonian terms as candidate cures, occupancy and bifurcation. The next two sections take them in turn; neither behaves the way its classical reputation suggests, and neither can be judged separately from the refit.

7 The occupancy term

7.1 The term

Denby's original network [5] constrains each hit to be used by at most one selected segment per role. The quadratic form of that constraint, per-hit *and per-role* occupancy (a hit's departing and its arriving segments are counted separately), adds $E_{\text{occ}} = \alpha \sum_h \sum_{r \in \{\text{in},\,\text{out}\}} (o_h^r - 1)^2$ with o_h^r the summed activation of the segments using hit h in role r, which at the matrix level reads

$$A_{\text{occ}} = A + 4\alpha I + 2\alpha B_{\text{all}}, \qquad \mathbf{b} \to \mathbf{b} + 4\alpha \mathbf{1}, \tag{16}$$

where B_{all} couples every pair of segments sharing a hit in the same role: two segments departing the same hit, or two arriving at it. Consecutive segments of one track share their interior hit in

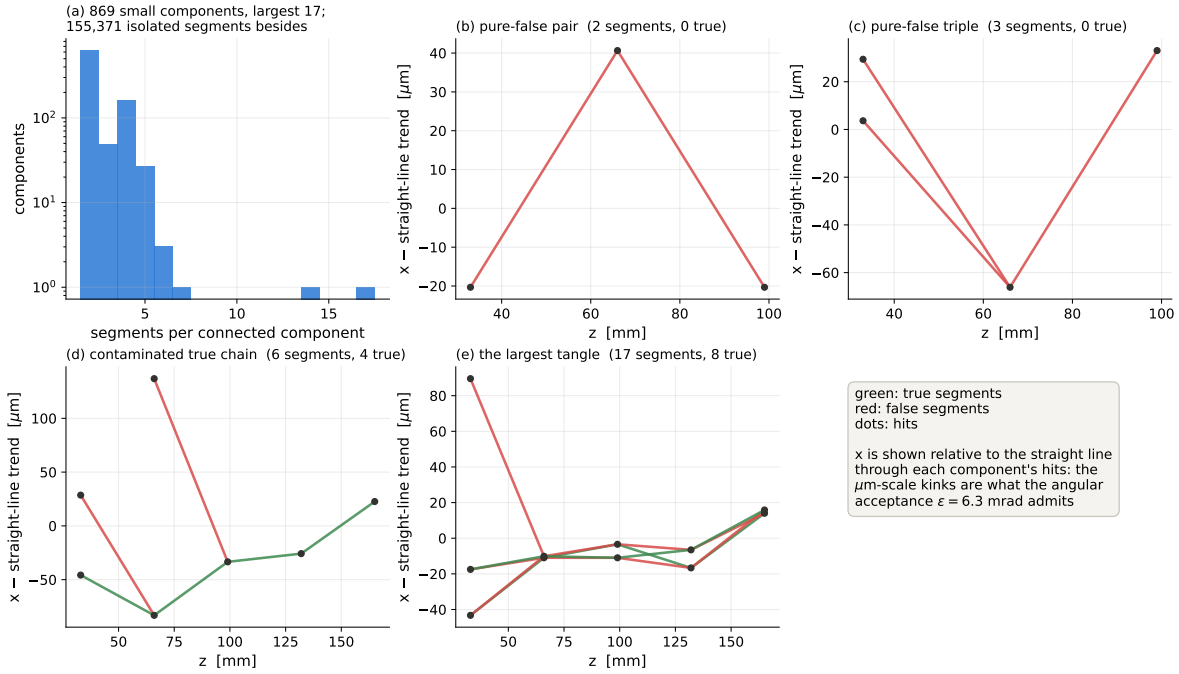


Figure 11: The heavy-noise compatibility graph, seen directly (heavy $T = 200$, rep 0). (a) Component-size distribution: many small motifs, no giant component. (b–e) Real components drawn in the detector, with the straight-line trend subtracted so the micron-scale kinks are visible: a pure-false pair, a pure-false triple, a true chain with false contamination, and the largest tangle in the event (17 segments, 8 true). Green segments are true, red false.

opposite roles and are not coupled. On an isolated chain the attractor moves to $(\delta{+}4\alpha)/(\gamma{+}\delta{+}4\alpha)$; on an event, as shown below, it does not.

That role structure has an exact consequence worth deriving before any experiment: *on every pure chain the occupancy term is nothing but the substitution* $(s, \delta) \to (s{+}4\alpha,\ \delta{+}4\alpha)$. No same-role pair lives inside a chain, so restricted to a chain of m segments eq. (16) reads $(s + 4\alpha)I - P_m$ with bias $\delta + 4\alpha$, and the entire closed-form vocabulary of section 2.3 survives verbatim. The ladder of eq. (9) generalises with $\bar{x}' = (\delta + 4\alpha)/(s + 4\alpha - 2)$ and $r' + 1/r' = s + 4\alpha$:

$$x_j = \bar{x}' \left[1 - \frac{r'^{\,j} + r'^{\,m+1-j}}{1 + r'^{\,m+1}} \right], \tag{17}$$

so at $\alpha = 0.05$ the attractor sits at $1.2/4.2 = 0.2857$, the P_4 outer and inner levels rise from $4/11$, $5/11$ to 0.4052, 0.5017, and every chain eigenvalue line of figure 4 shifts rigidly by $+4\alpha$. On an isolated chain, that is. An event is not a collection of isolated chains: every hit carries $T - 1$ false candidate segments per role, so the same-role sums the term penalises are, to within 0.2%, independent of which hit one looks at, and the term's effect on the solution collapses to a mean field. Measured at $T = 200$ and $\alpha = 0.05$, every segment amplitude, true or false, is the base amplitude multiplied by one number, 0.1087 (its 5th–95th percentile range across segments is 0.1087–0.1088 on clean events and 0.108–0.110 under noise), and the isolated-false level sits at $(\delta + 4\alpha)/(s + 4\alpha + 4\alpha(T - 1)) = 0.027$, not at the chain-only 0.286. The atlas does not move as a block; it shrinks as a block. What the term changes is the *ranking* inside the shrunken atlas, through the small per-hit departures from the mean field, and that is what the efficiency-first working point reads: the false rate at 99% efficiency is unchanged on clean events (0.29% both ways) and falls from 44% to 35% at moderate and from 65% to 30% at heavy noise. Every occupancy result below is read at that working point and stands; only the closed-form ladder, exact for an isolated chain, must not be read as a prediction of event amplitudes.

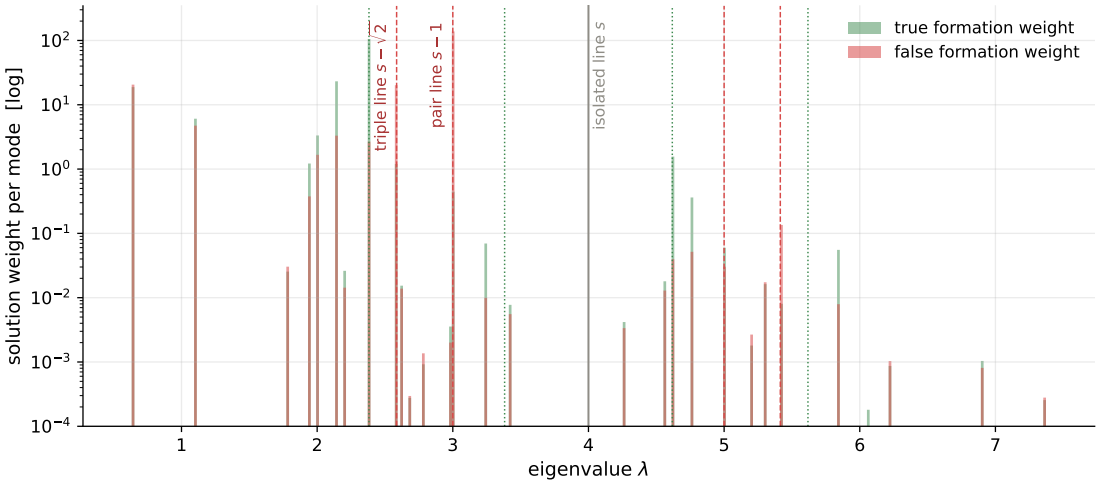


Figure 12: The measured spectral pile-up of the heavy $T = 200$ event (rep 0), computed exactly by per-component diagonalisation: solution weight per mode against λ, split into true (green) and false (red) content. The false weight concentrates on the pair line $s - 1$ and the triple lines $s \pm \sqrt{2}$ of the motif dictionary (figure 4), close to but never on the four true-track lines (green dotted).

What it does touch is exactly the structures the base Hamiltonian cannot see. A same-role pair with no continuation edge, invisible to A (both segments sit on the isolated line), acquires the 2α coupling and solves to $(\delta + 4\alpha)/(s + 6\alpha) = 0.2791$ at $\alpha = 0.05$ — *below* the shifted attractor by $\approx 2\alpha\delta/s^2$, with its degenerate line split to $s + 2\alpha$ and $s + 6\alpha$. This is the analytic seed of the floor-break measured in section 10: the floor theorem's invariant grows from the compatibility graph alone to the pair (C, B_{all}), so two structures are twins only if they agree in their chain shape *and* in their hits' same-role degrees — and the hit drop that creates a true fragment has just lowered those degrees. Isomorphism becomes rare instead of generic. The same algebra also prices the term's cost at scale: the same-role k-clique at a single hit contributes $k-1$ lines at $s + 2\alpha$ and one at $s + 4\alpha + 2\alpha(k-1)$, so the top of the spectrum grows linearly with the busiest hit's degree. That linear growth is the span wall that section 10 measures ($\lambda_{\max} \approx 143$ at T=700, 202 at T=1000), and it is why in-matrix occupancy will emerge there as a low-multiplicity technique.

7.2 Classically it works; the notch it destroys

On the heavy $T = 200$ benchmark event the in-matrix occupancy term at $\alpha = 0.3$ does exactly what Denby promised: the classical efficiency-first false rate falls from 0.650 to 0.210 at efficiency 0.991. If this work were classical, the section would end here with a recommendation.

For the 1BQF it is a disaster, and the mechanism is worth spelling out because it generalises. The notch's power is that isolated false segments sit exactly on its zero. The occupancy coupling $2\alpha B_{\text{all}}$ connects precisely those segments to their co-hit neighbours, which is its job, and thereby moves them off the zero. On the benchmark event the co-hit false bulk, notched to amplitudes $\sim 10^{-19}$ under the base Hamiltonian (99% of it below 10^{-17}), re-emerges under A_{occ} at amplitudes on the true band: its 99th percentile rises to 1.5×10^{-3} against a true median of 1.2×10^{-3}. The rank separation collapses (1BQF AUC 0.996 $\rightarrow$ 0.634) and no threshold rescues it; the full sweep shows every occupancy-1BQF operating point dominated by the base-1BQF curve. Nor does the self-coupling γ offer an escape: the notch tracks the diagonal, distance-to-notch is γ-invariant, and at the window-restoring $\gamma = 236$ the occupancy system is strictly worse. Post-selection tells the same story: the occupancy term diverts the ancilla-conditioned probability mass almost entirely onto false segments.

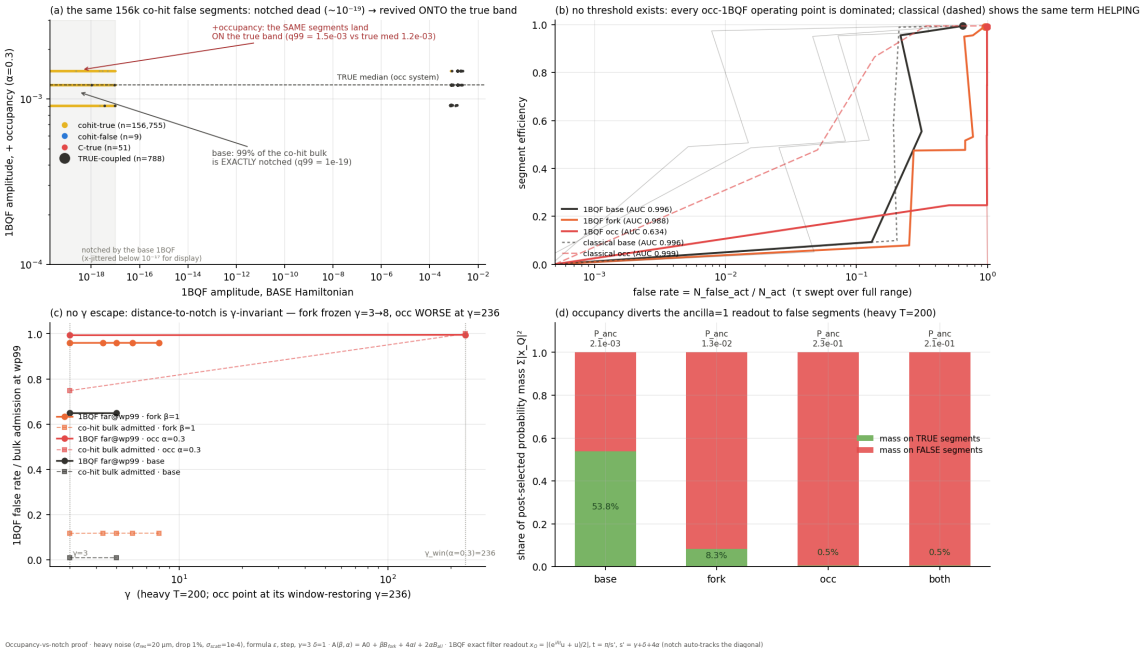


Figure 13: The occupancy proof (heavy $T = 200$): (a) the same co-hit false segments, exactly notched under the base Hamiltonian, re-emerge on the true amplitude band under A_{occ}; (b) no threshold escape: every occupancy-1BQF operating point is dominated, while the same term helps the classical solver; (c) no γ escape; (d) the post-selected probability mass diverts to false segments.

The mechanism generalises to any *fixed* response: a spectral filter earns its living from populations pinned to known eigenvalues, and any Hamiltonian term that couples a pinned population unpins it. For the 1BQF the verdict really is final, because the notch has no further freedom to spend: its response family is exhausted by the single parameter t, the retune $t = \pi/s'$ is already included above, and figure 13(c) shows the γ direction offers no escape either. As a Hamiltonian term, occupancy is poison to a notch, and the classical intuition for the term does not transfer to it.[7] But "final for the notch" is not "final for the filter family". The fork term in the next section will teach the general lesson of this work, that an operator modification can only be judged together with a response refitted to the spectrum it creates, and section 9.4 returns to the occupancy term with exactly that machinery. The verdict there reverses: what destroys the degree-one filter turns out to be the single most helpful operator modification in the study once the polynomial is allowed to follow it.

8 The fork term and the moving spectrum

8.1 The term and its validity envelope

The second Denby–Peterson penalty targets *bifurcations*: two segments claiming the same hit in the same role at small mutual angle, the fork topology that produces the closely-competing candidates a wide acceptance admits. The ε-windowed form adds

$$A' = A + \beta\, B_{\text{fork}}(\varepsilon_B), \tag{18}$$

[7] The occupancy constraint can also be applied entirely outside the Hamiltonian, as a classical hit-uniqueness post-selection applied evenly to every solver's output, deliberately kept out of this work, whose subject is the spectral side.

where B_{fork} couples co-hit same-role pairs within angular window ε_B (default: the acceptance itself), and the bias $\mathbf{b}$ is untouched. Unlike the occupancy term, the fork is surgical where it matters. On clean events it touches $\approx 2\%$ of segments, and provably no pure-true chain carries a fork edge — so not just the true comb lines but the *entire true-chain solution* is exactly β-invariant: the ladder of eq. (9) survives untouched, atlas levels included. What moves is precisely the topology the acceptance cannot resolve. An isolated competing pair (two same-role segments inside the window, no continuation edge) solves to $\delta/(s+\beta) = 0.222$ at $\beta = 0.5$, below the 0.250 attractor, with its degenerate line split to $s \pm \beta$: the fork prices exactly the closely-competing candidates it was built for. One caution comes from the same algebra: when a *false* prong forks onto a *true* chain's outer segment, the exact five-segment solve shows both ends move (the prong drops far below the attractor, the outer chain level dips with it), so the term's net effect on the true band is not sign-definite a priori — which is one more reason the fork can only be judged together with the response, the subject of section 8.2. Finally, $B_{\text{fork}} \subset B_{\text{all}}$, so under occupancy *and* fork a windowed pair carries the combined coupling $2\alpha + \beta$ and solves to $(\delta + 4\alpha)/(s + 6\alpha + \beta) = 0.250$ at the production knobs $(\alpha, \beta) = (0.05, 0.5)$ — numerically back at the base attractor, a coincidence of the operating point rather than anything deep. Earlier classical-side work in this programme established that the fork migrates contaminated-true spectral weight back toward the comb bands; whether real solvers can collect that migration is settled in section 9.

Adding a positive semi-definite coupling widens the spectrum, and the solver imposes two validity conditions with closed forms. Positive definiteness requires $\gamma > \gamma_{\text{pd}} = \gamma_{\text{ref}} - \lambda_{\min}$; the one-period condition of any cosine-family filter (the top of the spectrum must not alias through the response period) requires $\gamma > \gamma_{\text{win}} = \lambda_{\max} - \gamma_{\text{ref}} - 2\delta$, both extremes measured at a reference γ_{ref} and shifted rigidly, since γ enters only the diagonal. At heavy $T = 200$ the measured requirement is $\gamma^* \approx 3.4$ at $\beta = 0.5$ and ≈ 4.4 at $\beta = 1$. The notch tracks the diagonal automatically ($t = \pi/s'$, $s' = \gamma + \delta$), and the comb is rebuilt on the measured shifted spectrum. Every fork result below carries these retunes; a window-violating control ($\beta = 0.5$ at $\gamma = 3$) is kept in the study to show they matter.

8.2 The operator and the response are one design

One methodological point, learnt the hard way, belongs here rather than in a results table. Retuning a *fixed* response to the fork system, tracking the diagonal, rebuilding the comb on the shifted domain, is not enough, because the retunes move the response's centre and domain but never ask where the false weight now lives. Evaluated that way, both Denby–Peterson terms read as regressions, and the reading is a statement about the mismatch between an old response and a new spectrum, not about the terms. The correct procedure, used for every operator result in this work, is to measure the modified system's false pile-up and refit the polynomial to it: change the Hamiltonian, and the activations and their spectral support move; when they move, the polynomial must be refitted to the moved spectrum. Carrying out that refit, for the fork and then for the occupancy term, is the subject of section 9. (A second, smaller point: at heavy noise the efficiency-first convention degenerates for an unrefitted comb, whose contaminated-true amplitudes collapse toward zero, so sections 9 onward compare solvers at matched efficiency instead.)

8.3 A Trotter aside that will matter on hardware

Everything in this work so far treats the evolution e^{iAt} as exact. On hardware it is not: a quantum computer realises e^{iAt} by a circuit, and the standard cheap construction, the *product formula* (or Trotterization), builds it from the pairwise pieces of the Hamiltonian. The idea is simple: A is a sum of many small commuting-or-not terms, one per coupled segment pair, and while $e^{i(X+Y)t}$ is hard, the product $e^{iXt}e^{iYt}$ is easy, one small gate per term. If X and Y

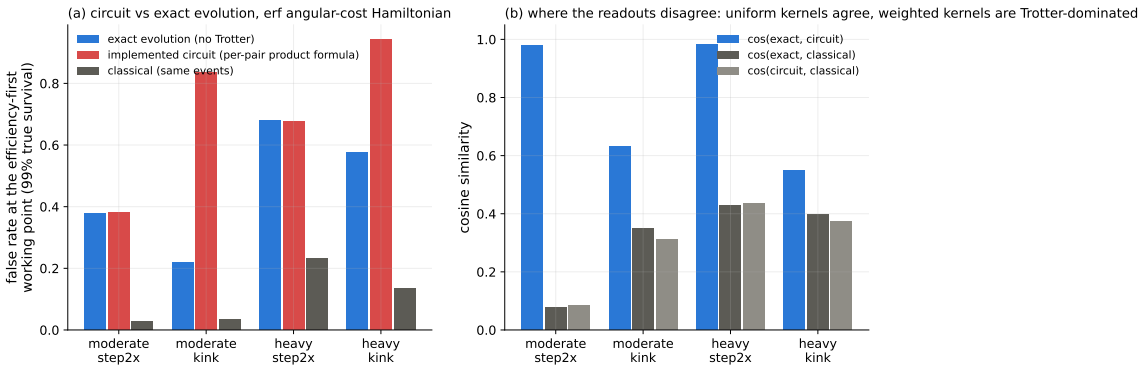


Figure 14: The exact-evolution arbiter on the erf angular-cost Hamiltonian (heavy and moderate noise, $T = 200$). (a) False rate at the efficiency-first working point: exact evolution against the implemented product-formula circuit and the classical baseline on the same events. Uniform-weight kernels (step2x) show circuit = exact, so the remaining deficit is physics; weighted kernels (kink) are Trotter-dominated. (b) The same story as readout fidelity: cosine similarity between the three solutions.

commute the product is exact; if not, it errs at order $t^2[X, Y]$, and one hopes the accumulated error stays below the physics. The 1BQF circuit of ref. [3] is exactly such a per-pair product.

The programme's measurements show where that hope holds and where it breaks, and the split is clean (figure 14). With *uniform* coupling weights (every term the same strength, as in the step-kernel Hamiltonian of this work) circuit and exact evolution agree to cosine similarity ≈ 0.985: the product formula is safe. With *heterogeneous* weights (the erf-weighted angular kernel, or a fractional fork $\beta = 0.5$ riding on unit compatibility edges) the per-pair product scatters the readout badly enough to dominate the physics: on the fork system the circuit reads false rate 0.96 where the exact evolution reads 0.73, and on the erf kernel removing the product formula alone collapses the false rate from 0.83 to 0.22 at moderate noise. The rule of thumb for hardware is that heterogeneous coupling weights break the naive product formula, and circuit realisations of weighted Hamiltonians need the structured synthesis route (edge-coloured pair ordering) developed in the programme's direct-structural-synthesis study. All fork and fit results in this work use exact evolution precisely so that filter physics and circuit error are never conflated; section 11.4 returns to what this means for deployment.

9 Fitting the polynomial to noisy events: roots at the measured pile-up

9.1 Measuring the false pile-up

The fit begins with a measurement no fixed design ever asked for: where, exactly, does the false amplitude come from? Because the heavy-noise compatibility graph has no giant component (section 6), the per-component eigendecomposition gives, for every eigenvalue in the event, its exact content: how much of the mode's weight lies on true segments, how much on false, and how much of each segment's amplitude it builds. Under any response p,

$$x_s(p) = \sum_i p(\lambda_i)\,(\mathbf{v}_i^\top \mathbf{b})\,v_i(s), \tag{19}$$

which is nothing more than the spectral decomposition of the filtered solution $p(A)\mathbf{b} = \sum_i p(\lambda_i)\,\mathbf{v}_i\mathbf{v}_i^\top\mathbf{b}$ read off at segment s. The two geometric factors are fixed by the event alone, before any response is chosen: $\mathbf{v}_i^\top\mathbf{b} = \delta\sum_s v_i(s)$ is the projection of mode i onto the uniform source (recall $\mathbf{b} = \delta\mathbf{1}$), and $v_i(s)$ is that mode's amplitude on segment s. The response enters

only through the scalars $p(\lambda_i)$. Three consequences follow, and the section rests on all three. First, each mode deposits a definite *formation weight* $(\mathbf{v}_i^\top \mathbf{b})\, v_i(s)$ onto every segment, so each segment's amplitude is a sum of contributions labelled by eigenvalue: a histogram of that weight against λ is the spectral pile-up of figure 12. Second, each mode carries a definite true/false *content*, the fraction $\sum_{s\,\in\,\text{true}} v_i(s)^2$ of its weight landing on true segments, so the false part of the amplitude can be attributed to specific eigenvalues rather than to the solver. Third, and decisively for what follows, x_s is *linear* in the values $\{p(\lambda_i)\}$: freeze the geometry, treat the response samples as the unknowns, and every segment amplitude is a fixed linear combination of them. The question of where the false amplitude comes from is therefore well posed and exactly answerable. The answer, on the heavy $T = 200$ events, is concentrated to a degree I did not expect. The false-active formation weight piles up on the *pair line* $\lambda = s - 1$: on the three events measured, that single bin carries 1122–1378 weight units against 4–10 units of true weight, with the triple lines $s \pm \sqrt{2}$ next; the overall bin-level overlap between false-active and true formation is 0.014–0.023 in cosine. The false population is not smeared. It is parked on the eigenvalue lines of identifiable two- and three-segment motifs, almost none of which are shared with true formation.

9.2 Fitting the response

Because eq. (19) is linear in the response, choosing the response is a least-squares problem. Represent p by its values on a grid of narrow bins in λ (width $\Delta\lambda = 0.02$, a few hundred bins spanning the spectrum); these binned values are the free parameters, and by eq. (19) each segment amplitude is a fixed linear combination of them, $x_s(p) = (D\mathbf{p})_s$ with D the design matrix that collects the formation weights into bins. The response is then the solution of

$$\min_{p} \underbrace{\sum_{s\,\in\,\text{false}} x_s(p)^2}_{\text{purity}} + \mu \underbrace{\sum_{s\,\in\,\text{true}} \left(x_s(p) - x_s^{\text{cls}}\right)^2}_{\text{efficiency}}, \qquad |p| \le 1. \tag{20}$$

The first term drives the false-segment amplitudes toward zero; the second pins the true-segment amplitudes to the values x_s^{cls} the classical inversion gives them, so the fit cannot manufacture a clean false spectrum simply by discarding true segments. Both terms are quadratic in the binned values, so the minimiser solves the linear normal equations

$$(D_F^\top D_F + \mu\, D_T^\top D_T)\, \mathbf{p} = \mu\, D_T^\top \mathbf{x}^{\text{cls}}, \tag{21}$$

with D_F, D_T the rows of D for false and true segments, and the $\mathcal{O}(10^5)$-strong isolated-false line at $\lambda = s$ carried into D_F as one heavily weighted row so it is held at the notch. The bound $|p| \le 1$ is the QSVT realizability constraint of section 4: only a response bounded by one on the spectrum can be implemented as a walk. The single knob is μ, which trades the two terms against each other: $\mu \to 0$ maximises purity at the cost of the true band, $\mu \to \infty$ recovers the classical efficiency at the cost of the false spectrum, and sweeping it traces the efficiency–purity frontier the results below report. No root position is chosen by hand: zeros appear where false weight concentrates because that is what the objective rewards. The fitted response reproduces the comb's passes at the (contamination-shifted) true lines and drives deep roots through $s - 1$ and $s \pm \sqrt{2}$: precisely the polynomial that no clean-event design could have anticipated, because the root locations are properties of the measured noise.

9.3 Results

Table 1 is the main result of the section, and it is worth being explicit about how to read it, because every number in it is the same kind of object. Fix a system (a row block: the base Hamiltonian, or the fork system at its validity point) and a target efficiency (the "eff" column).

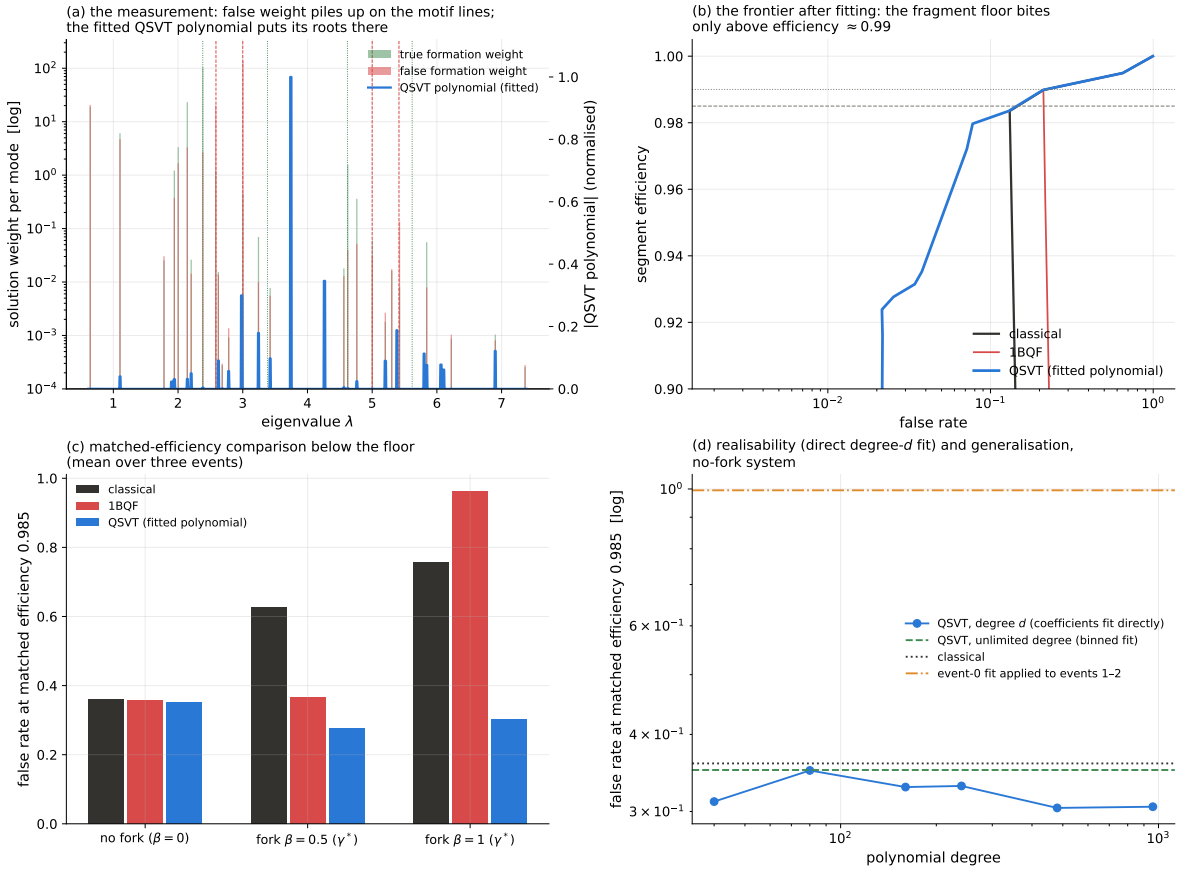


Figure 15: Fitting the response to the measured pile-up (heavy $T = 200$): (a) exact per-mode false (red) and true (green) formation weight against λ, with the fitted response overlaid; the fitted roots land on the pair and triple lines. (b) The resulting efficiency/false-rate frontier, with the twin fragments biting only above $\varepsilon_{\text{seg}} \approx 0.99$. (c) False rate at matched efficiency 0.985 for the base and fork systems (three-event means). (d) The same quantity for responses fitted directly at fixed polynomial degree: production-scale degrees already realise the measured optimum (section 11.2).

For each solver, slide that solver's own threshold until exactly the target fraction of true segments survives, the matched-efficiency convention of section 2.4, and record the false rate of the selected sample at that threshold. Lower is better; solvers in the same row are compared at identical efficiency by construction, so the false rate is the *entire* difference between them. All numbers are means over three events, heavy noise, $T = 200$.

system	eff	classical	1BQF	**fitted response**
base ($\beta = 0$)	0.98	0.124	0.199	**0.096**
base ($\beta = 0$)	0.97	0.124	0.199	**0.061**
fork ($\beta = 0.5, \gamma^*$)	0.98	0.463	0.145	**0.085**
fork ($\beta = 0.5, \gamma^*$)	0.97	0.292	0.124	**0.046**
either	0.99	0.63–0.76	0.65–0.96	0.63–0.77

Table 1: Segment false rate at matched efficiency (lower is better): each solver's threshold is set so the stated fraction of true segments survives, and the false rate of the resulting selection is quoted. Heavy noise, $T = 200$, three-event means. "Fitted response" is the optimal binned response of eq. (20), refitted to each system. The last row is the efficiency ceiling of section 9.3: at $\varepsilon_{\text{seg}} = 0.99$ every solver floods, on the base and fork systems alike.

Three conclusions follow.

First, the expected ordering is restored. Below efficiency ≈ 0.985 the fitted quantum filter sits under the classical baseline everywhere: 0.096 against 0.124 at $\varepsilon_{\text{seg}} = 0.98$, 0.061 against 0.124 at $\varepsilon_{\text{seg}} = 0.97$. The claim this section set out to test, that the false rate should still be lower for the quantum filters once they are fitted, holds.

Second, the fork earns its keep, but only through the refit. On the fork system the fitted response reaches 0.046 at $\varepsilon_{\text{seg}} = 0.97$, beating the fitted no-fork system's 0.061, while the classical solver on the same fork system reads 0.292. The operator knob does exactly what the spectral-migration analysis promised, provided the response follows it: changing the Hamiltonian without refitting the polynomial is not a half-measure, it is worse than doing neither (section 8.2). The standard-notch 1BQF also benefits from the fork below the floor (0.145 against 0.199 at $\varepsilon_{\text{seg}} = 0.98$): the fork pushes coupled-false modes away from the passband even when the only zero sits at s'.

Third, the wall at $\varepsilon_{\text{seg}} \geq 0.99$ is real on these systems, and now has a name. At the strict 99% point everything, classical included, floods back to $f_{\text{seg}} \approx 0.65$. The per-component decomposition locates the cause exactly: about 2% of *true* segments, a couple of dozen per event, counted directly and consistently on all three events, are spectral twins of the false motifs, in exactly the sense of section 5.5 and figure 10. They are the hit-drop fragments of section 5.3: a broken track's two-segment piece *is* a pair, sitting on the pair line, in a pure-true component, plus a handful of contaminated mixed chains. Any response with a root at $s - 1$ kills them alongside the false pairs; their population exceeds the 1% efficiency budget; and once the working point must dip below their suppressed amplitudes, the entire suppressed false band comes back above threshold. This is the *efficiency ceiling set by twin fragments*: the noisy-event generalisation of the clean floor theorem, no longer a curiosity but the binding constraint. It is a ceiling because it caps how far the efficiency can be pushed, and it is set by twins because the fragments and the false pairs it confuses are, to any function of A, the same object. The robust statement is the measured wall itself: on the base and fork systems every solver holds $\varepsilon_{\text{seg}} = 0.98$ and none holds $\varepsilon_{\text{seg}} = 0.99$, on all three events. On *those* Hamiltonians no function of λ can cross it, but the floor is a twin degeneracy of the operator, not of the event, and the next subsection shows an operator that removes it.

9.4 The occupancy term, refitted

Section 7 ended with an issue: the occupancy was measured on the fixed 1BQF targeting the central, isolated, eigenvalue, a filter with no freedom to follow the spectrum the term creates, and the fork has since taught us that operator terms can only be judged with a refitted response. The occupancy system makes the refit technically harder, the co-hit coupling $2\alpha B_{\text{all}}$ merges the

compatibility graph into a giant component, so the exact per-component measurement behind eq. (20) is unavailable, but not impossible: a degree-d response needs no eigendecomposition, because the moment vectors $T_k(X)\mathbf{b}$ are computable by sparse matrix–vector recursion and the amplitudes are linear in the Chebyshev coefficients. The fit of eq. (20) then becomes an exact least-squares problem in $d{+}1$ coefficients, over *every* segment of the event including the isolated bulk. Run identically on the base and occupancy systems (same algorithms, same events, only the Hamiltonian differs), it settles the question.

The result of section 7 contradicts (figure 16). At matched efficiency 0.98 the directly fitted response reads 0.085 on the base system and 0.050 on the occupancy system at $\alpha = 0.05$; at 0.97, 0.071 against 0.039. The term that destroys the notch roughly halves the false rate of the fitted polynomial, while the classical solver on the same systems sits at 0.12 throughout. Two structural findings come with the headline. First, *small coupling wins*: $\alpha = 0.05$ beats the classically-motivated $\alpha = 0.3$ (0.050 vs 0.067 at matched efficiency 0.98), because the occupancy term stretches the spectral span ($6.7 \to 43 \to 242$ for $\alpha = 0 \to 0.05 \to 0.3$) and a fixed degree budget loses resolution proportionally; the spectral benefit saturates long before the classical amplitude benefit does. Second, and most strikingly, *the occupancy coupling breaks that ceiling.* One of the three events is held at the ceiling on the base system, its fitted response cannot do better than 0.654 at matched efficiency 0.985, because its fragment twins are exactly degenerate with false pairs, and on the occupancy system the same event fits to 0.062. The mechanism is the floor theorem read in reverse: the twin argument requires isomorphic components, and the co-hit coupling embeds every motif in its hit-sharing neighbourhood, which is different for a fragment of a real track than for an accidental false pair. The embedding breaks the isomorphism, the degeneracy lifts, and a response fitted to the new spectrum separates what was provably inseparable on the base Hamiltonian. The ceiling of section 9.3 is therefore a property of the *base* Hamiltonian, not of the problem.

One engineering note: the degree budget that collects the occupancy win is larger than the base system's (the span costs resolution; most of the gain arrives by degree 80–160 at $\alpha = 0.05$). Like every fitted result in this section, the fit itself is mechanical, the roots land on the enumerable motif structure the operator produces, so the reversal is a property of the Hamiltonian, not of a tuning; folding it into the single fixed response of section 11.3 is part of that enumeration.

9.5 Why the 1BQF specifically cannot follow

A degree-one filter owns one zero, and the measurement says the heavy false spectrum has two mandatory root locations: the isolated line at s ($\mathcal{O}(10^5)$ segments) and the pair line at $s - 1$ (the active pile-up). The study's control makes the conflict vivid. Retuning the notch onto the pair line ($t = \pi/(s{-}1)$) does root the pile-up, and releases the isolated bulk, which re-enters at amplitude $|\cos(st/2)|\,\delta = 0.5$, on top of the true band, for a false rate of 0.995. The 1BQF's zero is spoken for. Covering both lines is degree two; covering the triple lines as well is degree four and up; and that, stated as a root budget, is in my opinion the cleanest justification for the whole QSVT programme that the toy has produced.

10 The operator axis at scale: degree, density, and the loss budget

The preceding sections established the method — measure the modified spectrum, refit the response — one operator at a time. This section reports the systematic campaign that crosses the two axes: every operator (base A, occupancy $\alpha = 0.05$, fork $\beta = 0.5$, and the combination) against every polynomial degree ($d = 4$ to 160, response refitted at each), at all three noise configurations (clean, moderate, heavy; formula ε throughout), on the same stored events with

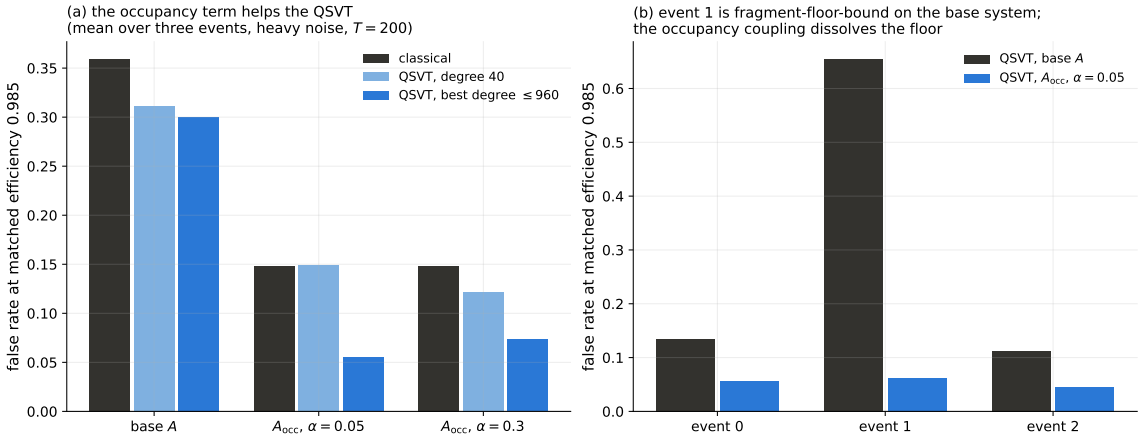


Figure 16: The occupancy term, refitted (heavy $T = 200$, degree-d responses fitted directly in coefficient space; identical machinery on every system). (a) False rate at matched efficiency 0.985, three-event means: the fitted response on the occupancy system at $\alpha = 0.05$ beats both the classical solver and the fitted response on the base system. (b) Per event: the event that is held at the efficiency ceiling on the base system (event 1) escapes it on the occupancy system: the co-hit coupling breaks the fragment/false-pair twin degeneracy.

the same matched-efficiency working points, ten event repetitions per cell. The classical solver and a matched 1BQF — the cosine $|\cos(\pi\lambda/2s')|$ realised as a degree-60 Chebyshev on the same operator — are computed on the same events, so every comparison in this section is like for like.[8]

10.1 The degree axis: what the polynomial can and cannot buy

Figures 17–19 answer the degree question in full, and the answer has a shape: a sigmoid in $\log d$, saturating completely. On moderate noise the saturation is already complete at $d{\approx}8$ for every operator — raising the degree from 8 to 160, a twenty-fold increase in circuit depth, buys a fraction of a percent of efficiency. On heavy noise the sigmoid is resolved: each operator has an onset degree below which the fitted response cannot place its roots and passes ($d{\approx}16$ for the fork system, 20 for base A, 64 and beyond for the occupancy operators), a steep rise, and a plateau (0.928 base, 0.938 fork, 0.935 occupancy at $d{=}160$) that no further degree improves. The marginal-gain plot (figure 19) is the operational summary: past the onset, one more degree step is worth nothing, and the budget question *which degree* reduces to *which operator*.

Two reference points sharpen this. First, the matched 1BQF: on base A it reads 1.000 (clean), 0.488 (moderate), 0.004 (heavy) at the 1% budget — and exactly 0.000 on *every* occupancy operator at *every* noise level, clean included. The degree-one filter does not merely degrade under the occupancy spectrum, as section 7 showed; it dies entirely, while the refitted $d{=}4$ polynomial on the same operator keeps clean events perfect. Second, the *unfitted* production comb run over the same grid: on the occupancy operators it reads 0.487 (moderate) and 0.000–0.004 (heavy) where the fitted response reads 0.990 and 0.935. A fixed response on a moved spectrum is not a weaker version of the method; it is a different, failing experiment — the strongest quantitative form of section 8.2's doctrine that the operator and the response are one design.

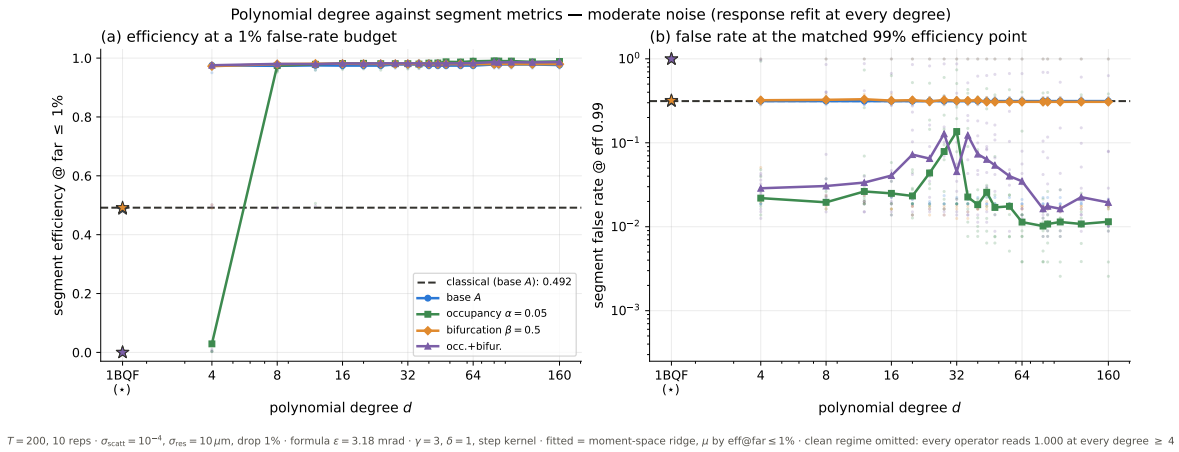


Figure 17: Polynomial degree against the two segment metrics on moderate noise (T=200, ten repetitions; per-repetition dots, median lines; response refitted at every degree). Stars at the left edge: the matched 1BQF, a degree-one filter, on each operator. (a) Efficiency at a 1% false-rate budget: every operator is flat from $d{\approx}8$ — the response axis is exhausted almost immediately, at 0.975–0.978 on base A against the classical 0.492. (b) False rate at the matched 99%-efficiency point: base A and the fork sit pinned at the 0.31 twin flood at *every* degree, while the occupancy operators alone drop to the 10^{-2} scale — purity at high efficiency is bought by the operator, not the polynomial.

10.2 The refitted responses, seen against the moved lines

Figure 20 draws the modified motif dictionary — every line position an exact closed form from sections 7 and 8 — and figure 21 shows what the fit does about it. The two figures carry the section's mechanism in miniature: the fork *sharpens* the spectrum (competing pairs leave the isolated line; the response collapses to one clean pass and its onset degree drops), while occupancy *stretches* it (the clique line drags the domain out to $\lambda \approx 44$ at moderate T=200, and the same degree spends its resolution over a window seven times wider). Both effects are visible in the degree scan of figure 18 as the ordering of the sigmoid onsets.

10.3 The loss budget: where the missing efficiency lives

Every true segment missed at an operating point is assigned to one of three named channels: *twin loss* — the segment's filtered amplitude agrees with a false segment's under *every* member of a set of independent probe responses, which by the floor theorem of section 5.5 certifies graph isomorphism; *ripple loss* — recovered by another degree of the same family; *threshold-overlap loss* — the remainder. The channels sum to the total loss by construction, and figure 22 shows the result. On base A at moderate noise the best fitted response reaches 0.974 at the 1% budget with zero ripple loss and zero threshold-overlap loss: *the deficit is one hundred percent twins*, and no function of λ on the unmodified operator can do better — the honest-negative bound of section 9 promoted from a scan result to a proof. The operator modifications act on exactly this channel. The twin fraction of true segments falls from 3.6% (base) to 0.76% under occupancy at moderate noise, and — the decisive number — from 87% to 1.3% at heavy noise, where on base A the twin flood forces a false rate of 0.633 at the 99%-efficiency point for *every* solver, classical included. The no-go knobs earn their verdicts in the same currency: shrinking the acceptance (ε-scale 1.5) *creates* twins (11% of trues) by stripping couplings, and moving γ moves the floor

[8] All numbers in this section derive from `QSVT/Efficiency_Frontier/` outputs; the degree scan is `06_degree_scan.csv`, the density scan `03_highT_frontier.csv` + `06b`, the loss budget `05_loss_budget.csv`. Classical post-processing (the hit-uniqueness gate studied elsewhere in this programme) is out of scope: every number is a property of the operator and the function applied to it.

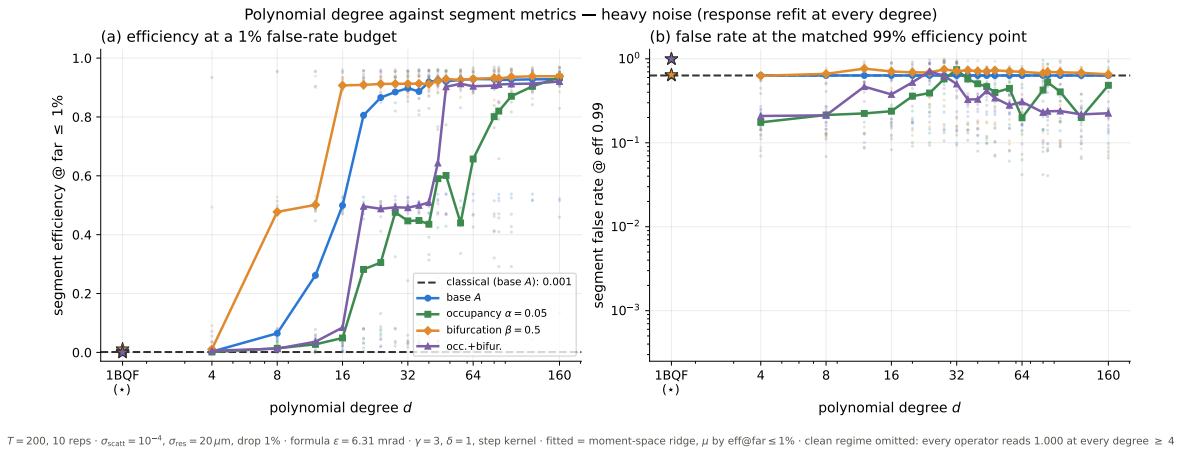


Figure 18: The same scan on heavy noise. The efficiency curves are sigmoids in $\log d$ whose onset degree is set by the operator: the fork rises first (0.91 by d=16), base A next (d≈20), the occupancy operators last ($d \gtrsim 64$), because the co-hit term widens the spectral domain and a fixed degree loses resolution proportionally (the span wall of section 10.4). The 1BQF stars sit at zero for every operator here.

the wrong way. Section 7's closed forms say why the occupancy term is the twin-breaker: the invariant that twins must match grew from the compatibility graph to the pair (C, B_{all}).

10.4 Density: every solver to $T = 1000$

Figure 23 carries the campaign to $T = 1000$ at the formula acceptance. Three structural statements survive to that scale. First, *clean events are solved at every multiplicity*: the degree-40 QSVT, coefficients refit, holds 99.85–100% efficiency at zero measured false rate to four million segments, on every Hamiltonian — while the 1BQF's false rate climbs through 0.10 (T=400) to 0.68 (T=1000) on the same events. Figure 24 puts the same solvers, at the same working point, back on the wide 2 mrad benchmark events of section 5: there the clean-event line-comb coefficients lose to the classical solver from $T \approx 550$, and the refit on each Hamiltonian reads 2.5–5.7% at $T = 1000$ against the classical 10.0%. The acceptance, not the solver, is what separates "solved outright" from a factor of three or four. Second, under noise there is a *composition wall*: the false-to-true ratio grows $\propto T$, and between T=700 and T=1000 the moderate-noise 1%-budget frontier closes for every operator alike; at the flooded matched-99%-efficiency points all five series read the same number — the false prevalence — which is the solver-independence of a completed twin flood, not a tie. Third, the occupancy term's floor-break *does not scale*: its spectral span grows linearly with the busiest hit's degree (section 7; measured $\lambda_{\max} \approx 143$ at T=700, 202 at T=1000), so at fixed degree the refitted response loses resolution and the moderate advantage decays from 0.9905 at T=200 through 0.952 at T=400 to nothing by T=700. In-matrix occupancy, as measured, is a low-multiplicity technique; the win it demonstrates at T=200 is a statement about what breaking the twin invariant is worth, not a recipe that survives density.

10.5 The two-axis rule

Table 2 is the study in four lines. Reading down the first column, the polynomial buys efficiency at a fixed false-rate budget: 0.49 → 0.978 with no change to the operator. Reading down the second, the response buys *nothing* at the matched high-efficiency point — classical, 1BQF and the best fitted polynomial all pay the identical 0.314, the twin flood of figure 22, exactly as the twin-loss budget predicts — and only the operator moves that number, thirtyfold, to 0.010. At heavy noise the same split holds in weaker form, and the combination earns its place:

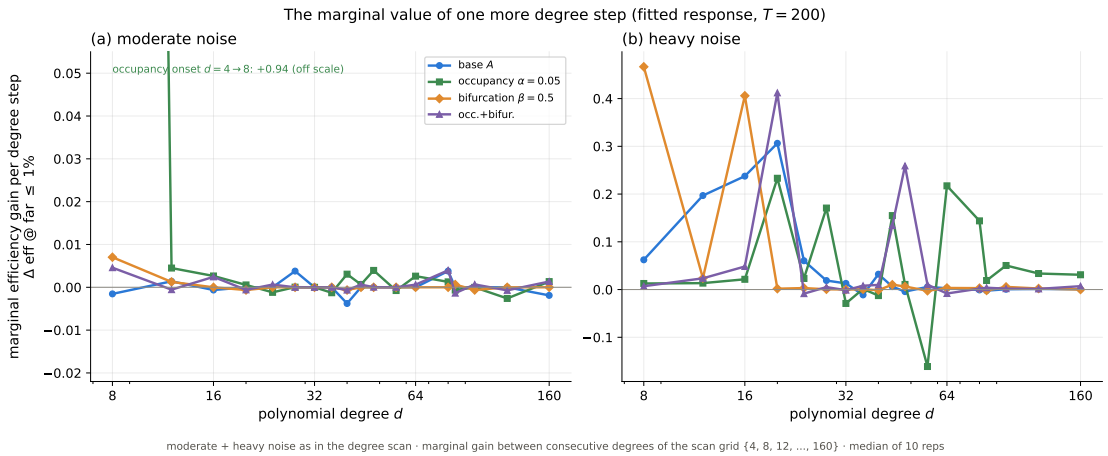


Figure 19: The marginal value of one more degree step (the derivative of the sigmoids of figures 17–18). (a) Moderate: after occupancy's onset step (+0.94 from d=4 to 8, off scale) every operator's marginal gain is consistent with zero at every degree. (b) Heavy: the marginal gain peaks at the operator's onset and decays to zero past it, with the non-monotone ripple of the fitted family visible as negative excursions.

Table 2: The two axes separated (moderate noise, T=200, medians of ten repetitions; every solver on the same events). Efficiency is quoted at a 1% false-rate budget; the false rate at the matched 99%-efficiency point. Fitted responses at their best scanned degree (base d=32, occupancy d=80).

solver / operator	eff. @ far $\leq$ 1%	far @ eff. 0.99
classical inversion, base A	0.492	0.314
1BQF, base A	0.491	0.314
QSVT fitted, base A	0.978	0.314
QSVT fitted, +occupancy	**0.9905**	**0.010**

occupancy+fork gives the best false rate at the 99%-efficiency point (0.225, against 0.356 for occupancy alone and 0.633 for base A). The two-part design rule of this paper's abstract is these two columns.

11 Limitations

I collect here, deliberately and in one place, everything that stands between the results above and a deployable algorithm. Each limitation comes with its measured size and, where we have one, its escape route.

11.1 The occupancy win does not survive density

The strongest configuration of section 10 — the occupancy operator with its refitted response — is, as measured, a low-multiplicity technique. Its spectral span grows linearly with the busiest hit's degree (the k-clique line of section 7), so a fixed-degree response loses resolution $\propto$ span/d and the moderate-noise advantage decays from 0.9905 at T=200 to nothing by T=700. What the T=200 result establishes is the *value* of breaking the twin invariant, not a recipe that scales. The untested escape routes are a multiplicity-dependent coupling ($\alpha \propto 1/\deg_{\max}$, pinning the span), a degree-normalised form of the co-hit term, and any operator that distinguishes fragments from false motifs without coupling entire co-hit cliques. Separately, the erf-kernel operator leaves the one piece of pure response headroom in the study: its twin floor is the smallest measured (0.25%

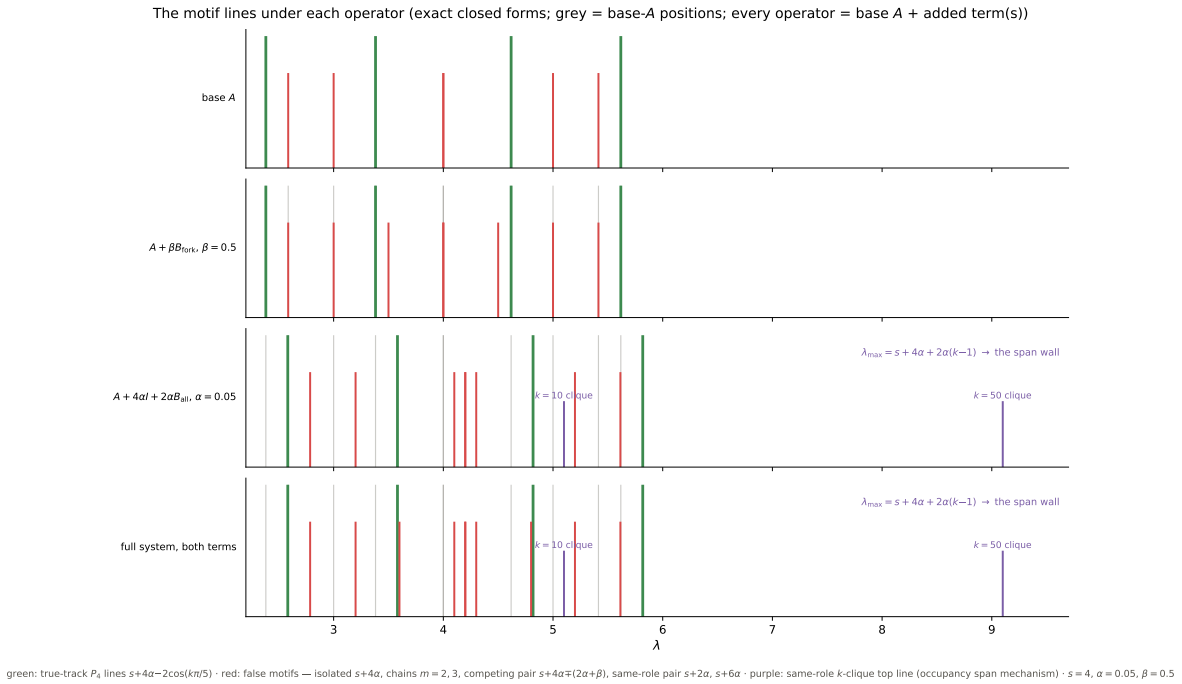


Figure 20: The motif dictionary of figure 4 under each operator, from the closed forms of sections 7–8 (grey: base-A positions). The fork moves only the competing-pair lines ($s \pm \beta$); occupancy shifts the true chains rigidly by $+4\alpha$ and splits the same-role pair to $s + 2\alpha$, $s + 6\alpha$; the same-role k-clique line $s + 4\alpha + 2\alpha(k-1)$ (purple) grows linearly with hit degree and sets the spectral span.

at moderate noise) but its loss is threshold overlap (1.3%), the one channel a better-shaped response can still buy back.

11.2 Realizability of the fitted response

An earlier version of this study believed realizability was the bottleneck, and the false alarm is worth recording because the resolution strengthens the result. The binned optimum of eq. (20) has features of width ≈ 0.02 on a spectral span of ≈ 7; a Chebyshev series *reproducing that specific function* needs degree $d \approx 2W/h \approx 700$, and truncated series fitted to it at $d \leq 240$ performed terribly. But reproducing the binned optimum is the wrong problem. Fitting the objective of eq. (20) *directly in the degree-d coefficient space*, amplitudes are linear in the Chebyshev coefficients, so this is the same least-squares problem in $d+1$ unknowns, finds realisable responses that match the unlimited-degree optimum already at the production degree (figure 15d: 0.099 at $d = 40$ against the binned optimum's 0.096 at matched efficiency 0.98 on the base system). The measured discrimination of section 9 is therefore available at circuit-affordable degree; nothing about the polynomial family was ever in the way, only the indirect two-step fit.

A structural escape remains available on top: the same no-giant-component fact that made the spectrum measurable makes the problem block-diagonal. Solving per connected component (a classical $\mathcal{O}(n)$ preprocessing step) replaces one $4T^2$-dimensional filter problem by hundreds of ≤ 17-dimensional ones, on which a degree-≤ 17 interpolating polynomial reproduces any response exactly, and as a bonus the register shrinks to $\lceil \log_2 17 \rceil + 2$ qubits and the success probability per cluster becomes $\mathcal{O}(1)$. The cost is honesty about what remains quantum: per-cluster solving helps the classical solver identically, and the quantum claim reduces to the resource profile of section 4.3 on whatever cluster sizes survive preprocessing.

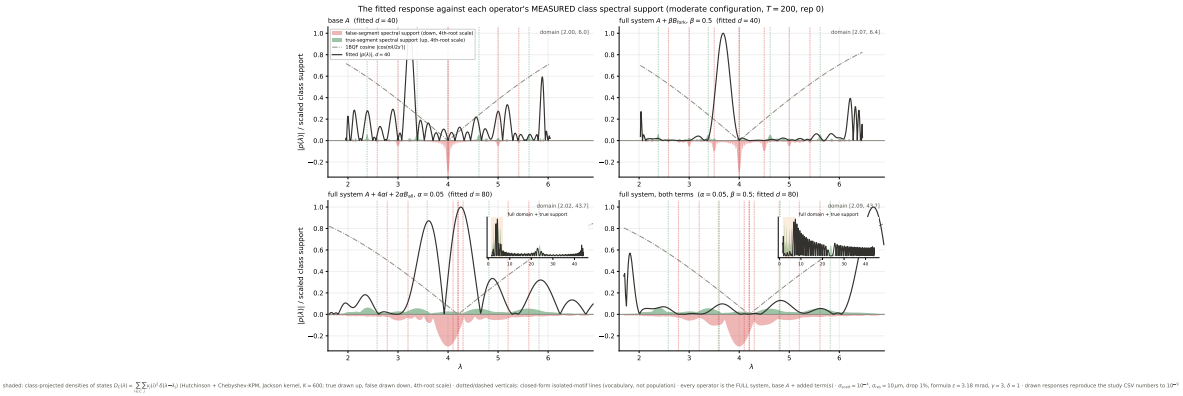


Figure 21: Figure 5's question asked of the fitted responses: $|p(\lambda)|$ per operator (rep-0 universal response, moderate noise) against that operator's exact motif lines. The fork system's response concentrates into a single dominant pass — the term has simplified the spectrum the fit must explain. The occupancy responses live on a span-widened domain (insets; the motif window is the shaded sliver), so their d=80 passes are visibly coarser inside it — the resolution cost span/d paid for breaking the twins.

11.3 One response for all events

On this toy the fit is not a statistical learning problem, and no train-and-test machinery is needed to understand it. The structures an event can contain are enumerable, chains, hubs, bridges and their contaminated combinations, each pinned to closed-form eigenvalue lines, and the fit is mechanical: the roots land on the motif lines because that is where the false weight must sit. The root positions are therefore universal ($s-1$ and $s\pm\sqrt{2}$ are motif properties, not event properties). What does vary between events is the contaminated-true pass positions, which chase each noise realisation's particular contaminated modes: a response taken verbatim from one event and applied to another reads 0.63–0.99 false rate against 0.04–0.10 on its own event, entirely because its narrow passes sit at the wrong contaminated positions. A production design is then an enumeration exercise, not a fitting one: roots pinned to the motif lines, passes wide enough to cover the contaminated-true spread of the ensemble. Building that single fixed response, and measuring what its wider passes cost against the per-event ceiling of section 9, is the programme's clear next step. On noisier data, real geometry, variable track length, denser contamination, the structure will no longer be enumerable by hand, and how to fix the response there will need closer thought (section 11.9).

11.4 Trotterization on hardware

Everything above uses exact evolution. On hardware, the 1BQF's product-formula circuit is exact only in the uniform-coupling case; the moment coupling weights differ (erf-weighted kernels, fractional fork strengths) the naive per-pair formula scatters the readout by amounts that dominate the physics (section 8). The structured-synthesis route (edge-coloured pair ordering, developed in the programme's DSS study, which also reaches 1BQF width for the comb via generalized QSP over e^{-iAt}) restores exactness at matched cost and is, on present evidence, mandatory for any weighted Hamiltonian on hardware.

11.5 Hardware resources: HHL, the 1BQF, and the QSVT side by side

The three algorithms of this lineage are best compared in a common currency, because on this problem they genuinely share one: all three are built from repeated calls to the same controlled time-evolution $e^{\pm iAt}$, synthesised gate-by-gate from the Hamiltonian's pair structure (the direct-structural-synthesis construction of ref. [3]: one Givens unit per coupled segment pair,

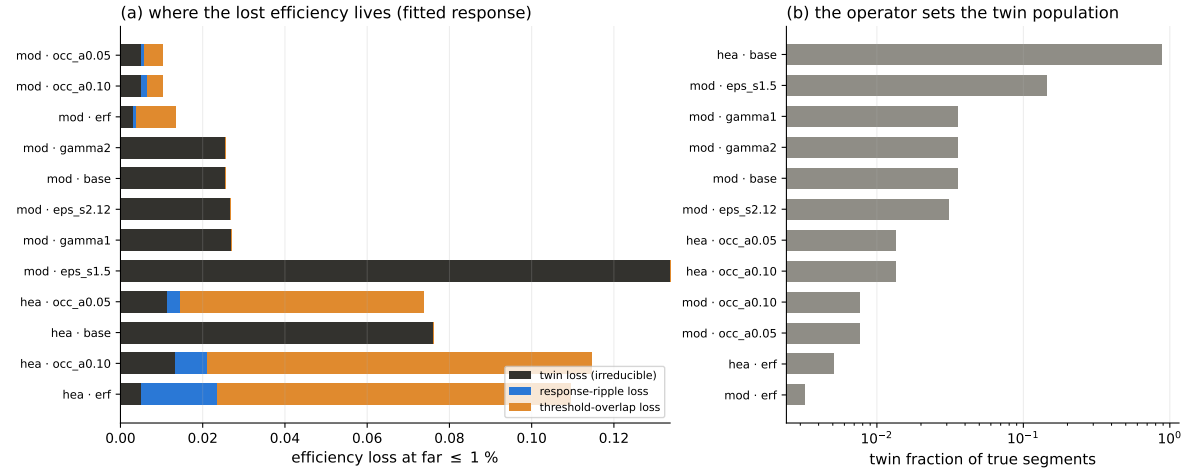


Figure 22: (a) The efficiency lost at the 1% false-rate budget, per operator (fitted response at its best degree), decomposed into its three named channels: twin loss, ripple loss, and threshold-overlap loss. On base A the entire deficit is twin loss — spectral twins, irreducible for any response. (b) The twin fraction of true segments per operator (log scale): the operator sets the twin population; the response cannot touch it.

a CNOT ladder plus one multi-controlled rotation, transpiled to a {CX, R_z, $\sqrt{X}$, X} basis). What distinguishes the algorithms is only *how many* calls they make: the 1BQF interferometer makes two; the degree-d comb, realised as a Fourier comb over the same primitive, makes d; and HHL wraps the evolution in phase estimation, making $\approx 2(2^{n_c}-1)$ calls for an n_c-bit clock register, an exponential price paid for amplitude information the selection task never uses, which was the founding observation of the 1BQF. Table 3 collects the measured numbers. Widths are exact register counts; two-qubit gate counts are measured on the stored events (the multi-controlled rotations transpiled, the CNOT ladders counted from the measured Hamming distances, $k \approx 1.5$–$2.3\sqrt{N}$ coupled pairs); the accompanying one-qubit $R_z/\sqrt{X}$ counts are of the same order as the CX counts (each ladder CX is dressed by $\mathcal{O}(1)$ single-qubit rotations) and are subdominant in every error budget, so CX is quoted. The HHL column is a structure-level estimate at $n_c = 6$–8 clock bits, the range a useful inversion on this spectrum would need; the predecessor implementations restricted the phase estimation precisely to tame this factor [1, 4]. The dense-dilation block encoding used by this work's own simulations is excluded from the table deliberately: it is simulation-grade only ($\approx$ 84k CX already at $T = 3$, with no structure for synthesis to exploit). Figure 25 shows how every row of the table scales with track count, measured from $T = 10$ to 1000.

	scaling with T	HHL	1BQF	comb (QSVT)
global event register				
logical qubits ($T = 1000$)	$\lceil \log_2 4T^2 \rceil + \text{const}$	$\approx$ 29–31	24	24
evolution calls per shot	constant	$\approx$ 130–510	2	40
CX per call ($T = 400$)	$\propto k\, n_s$; measured $\approx T^{1.4}$	6.2×10^5 (shared primitive)		
CX per shot ($T = 400$)	calls $\times$ per-call	$\sim 10^8$	1.2×10^6	2.5×10^7
amplification rounds ($T = 400$)	$\propto \sqrt{T}$	n/a	24	
per-cluster route (section 11.2, $\leq$ 17-dim blocks)				
logical qubits	independent of T	$\approx$ 12–14	6	6
evolution calls per shot	constant	$\approx$ 130–510	2	16
CX per shot (largest cluster)	set by cluster size, not T	$\sim 10^5$	$\approx 1.8 \times 10^3$	1.4×10^4
amplification rounds	constant	n/a	3	

Table 3: Logical resources of the three filters on the shared events, in the common currency of controlled-$e^{\pm iAt}$ calls (all three use the same synthesised evolution circuit; one call costs 6.2×10^5 CX at $T = 400$ globally, ≈ 900 CX on the largest 16-segment cluster). The scaling column gives the T-dependence of each row; anchors are quoted at the stated T. The HHL column is a structure-level estimate at $n_c = 6$–8 clock bits; the 1BQF and comb columns are measured. In the per-cluster route only the *number* of cluster instances grows with T (linearly); every per-instance resource is set by the cluster size, which the no-giant-component measurement caps at ≈ 17 segments. One-qubit gate counts track the CX counts at the same order and are omitted.

What error rates make each column viable? Run uncorrected, a circuit of N_2 two-qubit gates survives with probability $\approx (1 - p_2)^{N_2}$, so the usable budget is $N_2 \lesssim 1/p_2$. At today's best two-qubit fidelities ($p_2 \approx 10^{-3}$) that budget is $\sim 10^3$ gates: the per-cluster 1BQF ($\approx 2 \times 10^3$ CX on the largest cluster, far less on typical ones) sits exactly at this edge, which is why it is the member of the family that has run on real devices [3]. The per-cluster comb needs $p_2 \approx 10^{-4}$–10^{-5}, one to two orders beyond present hardware; every global column is out of uncorrected reach entirely.

Beyond that budget the requirement becomes fault tolerance, and the arithmetic is standard [11]. A distance-d_c surface code patch turns physical error p into logical error per operation $p_L \approx 0.1\,(p/p_{\text{th}})^{(d_c+1)/2}$ with threshold $p_{\text{th}} \approx 10^{-2}$, at a cost of $\approx 2d_c^2$ physical qubits per logical qubit. Demanding $p_L \lesssim 1/N_2$ at $p = 10^{-3}$: the per-cluster comb ($N_2 \approx 10^4$, 6 logical qubits) needs $d_c \approx 7$, i.e. $\sim 10^3$ physical qubits, a near-term fault-tolerant demonstrator. The global comb ($N_2 \approx 2.5 \times 10^7$, 24 logical qubits) needs $d_c \approx 13$, $\sim 10^4$ physical qubits; global HHL, another order deeper, needs $d_c \approx 15$ on a wider register. Because the code distance grows only logarithmically in depth, all the global routes land within a factor of a few of 10^4 physical qubits; the real separations between the columns are wall-clock time (the call counts) and the per-cluster row. Two honest caveats: these are logical-resource estimates from measured circuit counts and standard surface-code scaling, not a compilation study; and the Givens rotations carry arbitrary angles, so in a fault-tolerant run the dominant cost would shift from the CX counts quoted here to magic-state distillation for the rotation synthesis. The practical reading of the table is the same from either end: the per-cluster route of section 11.2 is where this family meets hardware first, the 1BQF already inside the uncorrected budget and the comb one hardware generation behind it.

11.6 A block encoding priced by hits, not matrix entries

One assumption sits underneath every gate count in table 3, and relaxing it is the most promising resource lever the programme has found. That assumption is how the Hamiltonian enters the circuit: the *block encoding*, the unitary whose top-left corner is A/α and on which every walk call

is built. The per-call cost in the table is priced the expensive way. The 6.2×10^5 CX evolution call carries one circuit element per matrix nonzero, and so does every literature block encoding the programme has benchmarked on these matrices (FABLE and its sparse variants, explicit sparse-oracle circuits [12], dictionary encodings): all of them pay $\Omega(T^2)$ gates per call, a circuit as large as the classical matrix. But A is a derived object. The event itself is $5T$ hits, $\mathcal{O}(T \log T)$ bits of information, and the matrix is a fixed geometric function of them: two segments couple exactly when they share a hit and their kink angle passes the acceptance of eq. (6). A block encoding that queries *hits* rather than matrix entries (a quantum read-only memory over the $5T$ sorted hit coordinates, fixed-point extrapolation arithmetic, and a comparator against ε, assembled in the standard state-preparation-pair form) costs $\mathcal{O}(T)$ gates per call instead of $\mathcal{O}(T^2)$. The programme's block-encoding study has validated this construction on the stored events: the hit-window enumeration reproduces every accepted coupling exactly (coverage 1.0 at every T from 5 to 1000, clean and noisy events alike), the subnormalization sits at the local-occupancy floor ($\alpha = 2$–4 up to $T = 1000$, the same class as the best entry-priced encodings, near the provable $\|C\|$ lower bound), and the modelled per-call cost at $T = 400$ is 1.6×10^4 CX against 2.4×10^6 for the entry-priced route, two orders of magnitude, and growing with T. Combined with the metric-validated comb degree (12–20 suffices on the base Hamiltonians; the production $d = 40$ is conservative), an entire hit-oracle comb solve costs 3.3×10^5 CX at $T = 400$: less than a single entry-priced evolution call in table 3. The price is width: the memory workspace adds ≈ 47–49 qubits over the 21-qubit native floor, so the offer on the table is qubits for depth: roughly 3–4× the register buys 4–7× less depth per solve at equal or better false rate. Two further measured facts complete the picture. The hit coordinates need 12 bits of fixed-point precision, and this is a hard cliff rather than a slope: at 10 bits enough ε-boundary couplings flip that the *classical* solve on the same quantized matrix breaks too, so the cliff is a property of the acceptance geometry, not of the encoding. And the erf-weighted kernel costs nothing extra, because the coupling weight is an arithmetic function of the kink angle the oracle already computes, where the entry-priced dictionary encoding sees its subnormalization explode on continuous values. None of this touches the floor theorem, which is representation-independent; it moves only the hardware accounting, and it moves it in the comb's favour.

11.7 Success probability and readout

The filtered state must still be post-selected and read out, and it is worth being explicit about what that costs, because it is the largest single overhead in the whole scheme (figure 26). The LCU post-selection succeeds when the index and dilation registers both return zero, which happens with probability P_{anc} of eq. (15); physically, P_{anc} is the fraction of the input state's norm that survives the filter. Because the uniform input spreads over all $4T^2$ segments while only $\mathcal{O}(T)$ of them carry surviving amplitude, P_{anc} falls as $1/T$: measured $\approx 0.44/T$ for the comb and estimated $\approx 0.25/T$ for the 1BQF, i.e. thousandths at $T = 1000$. Left alone this would mean thousands of repetitions per useful sample. Amplitude amplification (the standard Grover-type trick) reduces the repetition count from $1/P_{\text{anc}}$ to $1/\sqrt{P_{\text{anc}}}$, restoring a $\sqrt{T}$ scaling of total walk calls per useful sample, and active-set readout (sampling until the $\mathcal{O}(T)$ surviving segments are each seen) gives the measured $d \cdot T^{1.5} \log T$ walk-call budget against the classical $d \cdot T^2$ flop budget: a square-root-class separation for that readout task, and no more than that.

11.8 Classical simulability of the response family

Every response function in this work can be applied classically by the Chebyshev recursion in $\mathcal{O}(d \cdot \text{nnz}(A))$ flops, seconds at $T = 1000$. The discrimination physics (combs, roots, floors) is solver-independent mathematics of the segment Hamiltonian, and this work's contribution to it stands whether or not a quantum computer is ever involved. The quantum claim is confined to the resource profile: width logarithmic in problem size, walk calls $d\sqrt{T}$ against classical dT^2 for

the readout task above, and a hardware-native route to the same amplitudes. Readers allergic to quantum advantage claims may read the whole work as classical spectral-filter engineering with a quantum implementation appendix; nothing in the physics sections would change.

11.9 The toy's geometry, and what a real detector changes

Finally, the largest caveat of all: the four-line comb exists because this toy pins every true track to five hits. Real VELO tracks have variable length, and a companion study on real Run-3 event geometry found the true spectrum smeared into a band, the P_4 comb keeping only 40% of true segments, and a short fragments ($m \leq 2$ tracks indistinguishable from noise at the segment level) capping all segment-level solvers, while the fork transfer and the band-plus-notch redesign behaved as the toy predicts. Figure 27 shows the measured amplitude pile-up on the real geometry, side by side for the three solver designs: where the toy's amplitude atlas (figure 3) has discrete levels, the real geometry smears the true population into a continuum, the fragment population ($m \leq 2$, orange) sits inside the false bulk for every solver, and the redesigned band-plus-notch response reproduces the toy's qualitative separation on the findable ($m \geq 3$) population. The honest statement of scope is that this work's method is response engineering on whatever spectral structure the geometry provides, demonstrated exhaustively on a geometry that provides a lot of it; the measurement loop of section 9 (measure the pile-up, fit the response, locate the floor) is geometry-agnostic even where the specific comb is not.

12 Conclusion

This work set out to finish a thought that the one-bit quantum filter started. The 1BQF's insight was that track finding does not need a linear-system solution, only a spectral separation, and that one cosine with one zero buys the largest separation in the problem at almost no quantum cost. Promoting that cosine to the full QSVT polynomial family converts algorithm design into response design, and response design turns out to be a measurement-driven loop:

1. Read the spectrum the problem gives you. On clean events the detector geometry hands you four exact eigenvalue lines, and a forty-degree comb pinned to them beats the classical solver at the wide 2 mrad benchmark acceptance up to $T \approx 400$ tracks, every solver read at the threshold that keeps 99% of its true segments. Beyond that the clean-event coefficients must be refit to the events, and then the same polynomial converts the segment selection at $T = 1000$ from a 10.0% false rate (classical) to 2.8%: a factor of 3.6 fewer fakes in the selected sample. At the formula acceptance the same multiplicity is solved outright — 99.85% efficiency at zero measured false rate (section 10).

2. When the problem changes, measure where the false weight went, and put roots there. At the heavy-noise operating point ($\sigma_{\text{res}} = 20\,\mu$m, 1% hit drop, $T = 200$) the false weight piles up on the pair and triple motif lines, and the response refitted to that measurement reads a false rate of 0.061 at matched segment efficiency 0.97 where the classical solver on the same events reads 0.124, half the fakes at identical efficiency, on every event tested, at every efficiency below the twin-fragment ceiling.

3. Judge every Hamiltonian modification together with the response fitted to it, never separately. Both Denby–Peterson terms destroy the degree-one filter, and both help the refitted polynomial: the fork system fits to 0.046 at matched efficiency 0.97 (against 0.061 without the fork), and the occupancy system at small coupling $\alpha = 0.05$ fits to 0.039, the best number in the study, produced by the same term whose coupling collapses the 1BQF's separation (AUC 0.996 $\rightarrow$ 0.634). Operator and response are one design, not two.

4. Know what the spectrum cannot see, and know that the operator can change what that is.

Spectral twins bound every function of λ on a fixed Hamiltonian, classical scoring included; at heavy noise the true fragments of broken tracks are twins of the false pairs, and on the base and fork systems they floor every solver at segment efficiency ≈ 0.98: pushing to 0.99 floods the selection back to a ≈ 0.65 false rate for everything. But the floor is a degeneracy of the operator, not of the event: the occupancy coupling embeds each motif in its hit-sharing neighbourhood, breaks the twin isomorphism, and the one event that was floor-bound at 0.654 on the base system fits to 0.062 on the occupancy system at the same matched efficiency.

The open problems are, I think, unusually crisp for this field: a single fixed response for the whole event ensemble, on this toy an enumeration over the finite motif structure (roots on the motif lines, passes wide enough for the contaminated spread) rather than a fitting problem, with the floor-breaking occupancy system folded in; per-cluster realization as the structural alternative to global filtering; edge-coloured synthesis for weighted couplings on hardware; and the transplant of the whole loop onto real detector geometry, where the structure is no longer enumerable by hand, the response question needs closer thought, and the first measurements say the method survives even though the comb does not. Every cloud has a silver lining: the same noise that broke the clean-event comb handed us a false population parked on nameable eigenvalue lines, and a filter family with exactly enough roots to meet it.

Acknowledgements

To be completed

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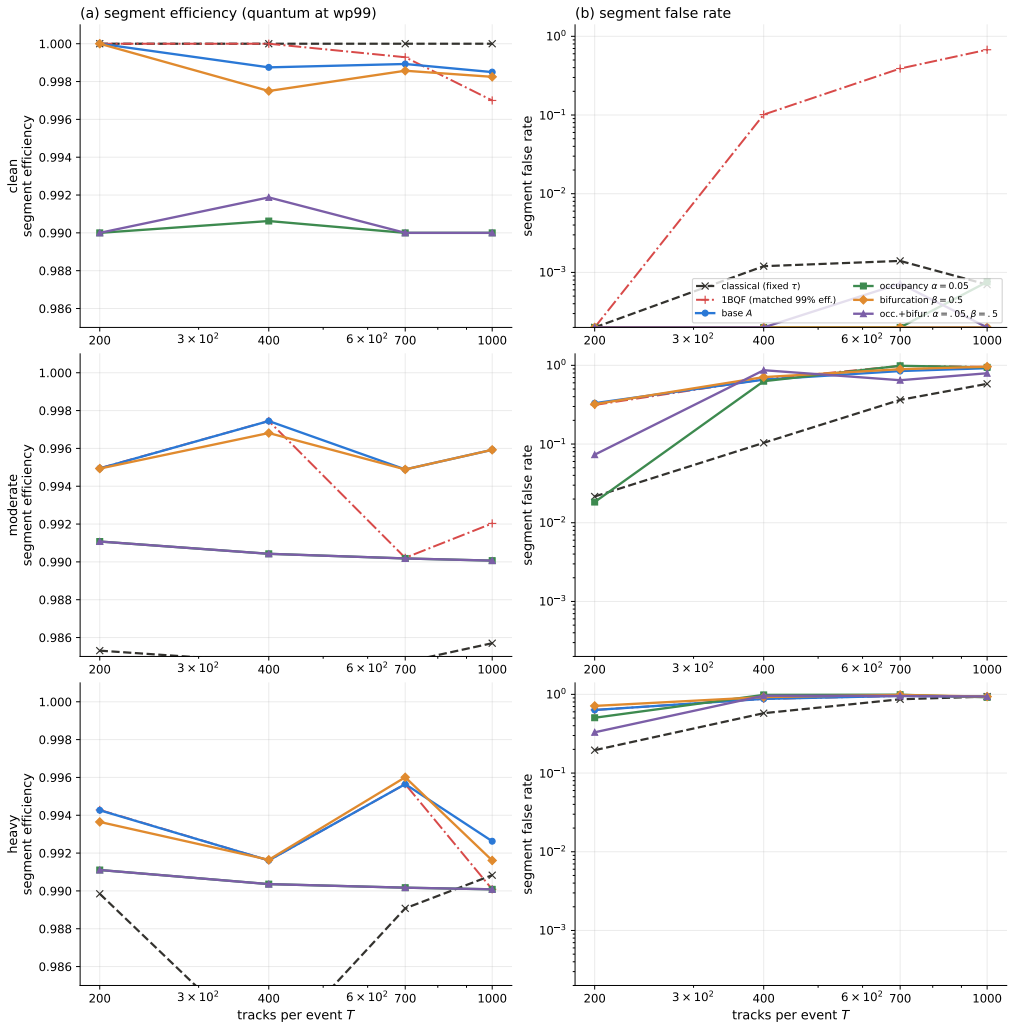


Figure 23: The figure 8 pair at the formula acceptance for every series — the classical solver, the 1BQF, and the degree-40 QSVT with coefficients refit on base A, A+occupancy, A+bifurcation, and both — at clean, moderate and heavy noise (rows), $T = 200$–1000, every solver at its efficiency-first working point. Clean is solved at every multiplicity on every Hamiltonian. Under noise, every curve converges to the false prevalence as the twin flood completes: past the wall the 99%-efficiency point stops being a working point at all. The classical solver, read at the same working point, sits at 31%–92% (moderate) and 63%–94% (heavy) false rate over this range.

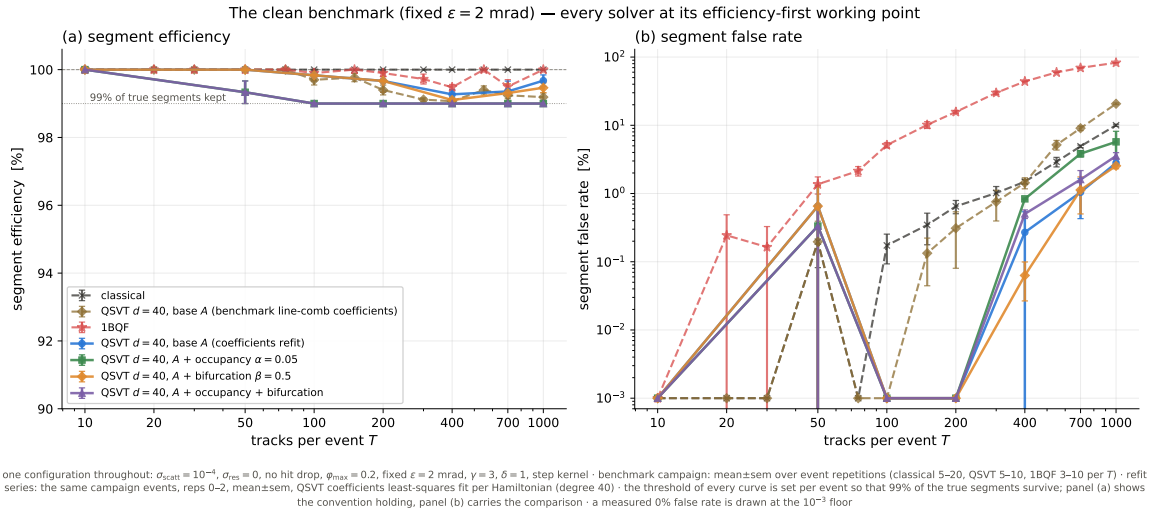


Figure 24: The clean benchmark of figure 8 (fixed $\varepsilon = 2\,\text{mrad}$, $\gamma = 3$, no hit drop) with every solver at its efficiency-first working point (99% of true segments kept; panel (a) shows the convention holding): the classical solver, the 1BQF, and the QSVT (degree 40, base A, the clean-event line-comb coefficients; benchmark campaign, mean ± s.e.m. over 3–20 repetitions), together with the same degree-40 QSVT with coefficients refit on base A, A+occupancy ($\alpha = 0.05$), A+bifurcation ($\beta = 0.5$) and both (repetitions 0–2 of the same events). The line-comb coefficients stop winning once the 2 mrad events develop tangles ($T \geq 400$); the refit restores the margin, 2.8% against the classical 10.0% at $T = 1000$, on every Hamiltonian. A measured 0% false rate is drawn at the 10^{-3} floor.

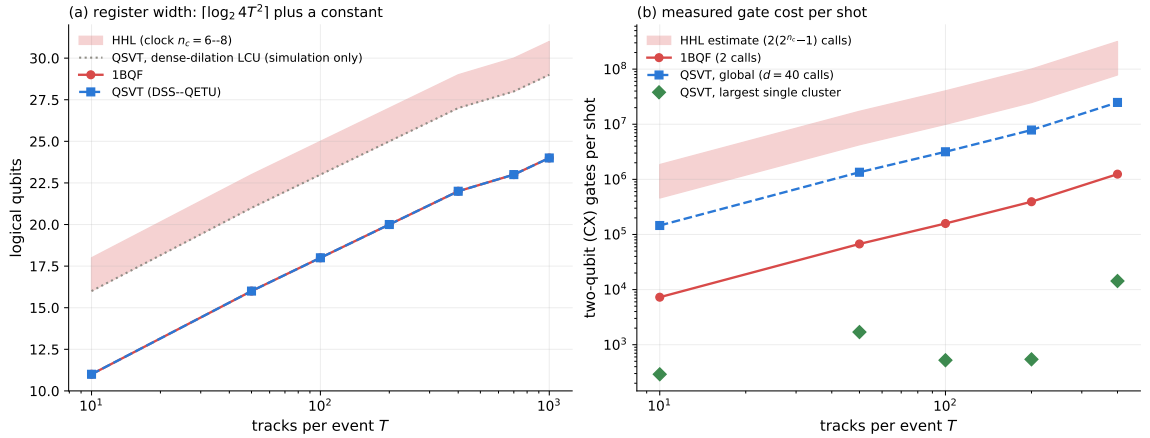


Figure 25: The rows of table 3 against track count. (a) Logical register width: all constructions follow $\lceil \log_2 4T^2 \rceil$ plus a constant: the 1BQF and DSS–QETU comb lines coincide exactly (the comb pays no width for its degree on this primitive), while the dense-dilation LCU pays +5 and the HHL clock band +7 to +9. (b) Measured two-qubit gate cost per shot: the shared evolution call grows as $\approx T^{1.4}$ (pair count × ladder length), the three algorithms sitting at fixed multiples of it (2, 40, ≈ 130–510 calls). The green diamonds are the largest single per-cluster instance at each T: set by the local cluster size, not by T, the flat exception to every other curve's growth.

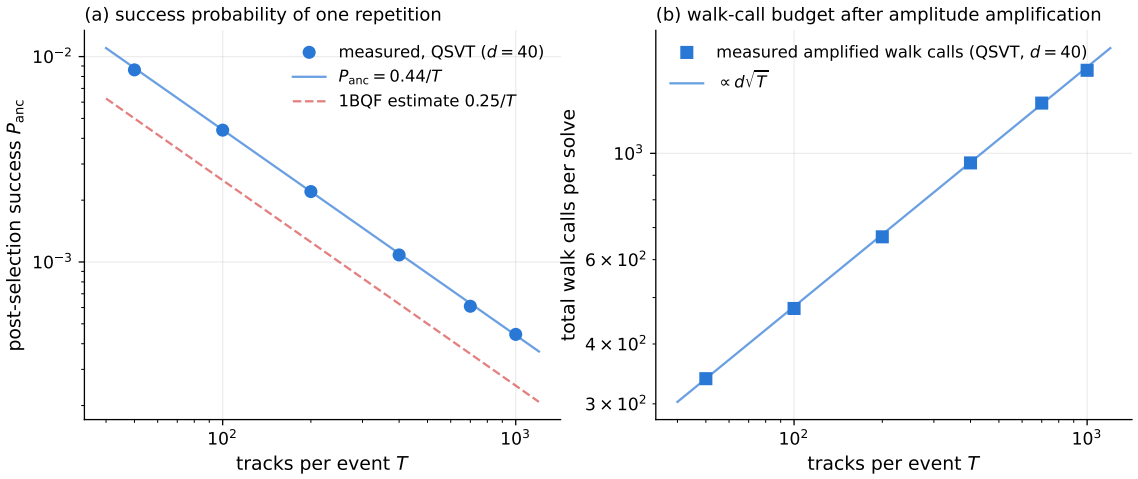


Figure 26: The readout budget, measured on the shared events at the production degree $d = 40$. (a) Post-selection success probability per repetition: the measured comb points follow $0.44/T$; the 1BQF estimate $0.25/T$ is shown for comparison. (b) Total walk calls per solve after amplitude amplification, following the $d\sqrt{T}$ scaling.

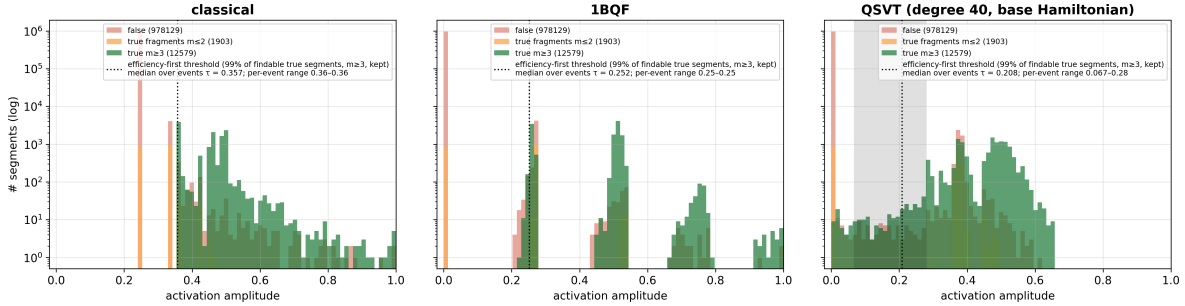


Figure 27: The real-geometry counterpart of the toy's amplitude pile-up: activation-amplitude spectra on real Run-3 VELO event geometry, for the classical solver, the 1BQF, and the redesigned band-plus-notch response. Pink: false segments; orange: true fragments ($m \leq 2$); green: findable true segments ($m \geq 3$); the dotted line is each solver's efficiency-first threshold, set so that 99% of the *findable* true segments (tracks with three or more true segments) survive, with the per-event range shaded; over all true segments the same rule is degenerate, because the 13% of true segments in fragments of one or two segments sit at the isolated level inside the false bulk. The QSVT here is the degree-40 band-limited design on the base Hamiltonian, not the line comb: real tracks have variable length, so the five-hit lines do not apply. Variable track length smears the discrete toy levels into bands, and the fragment population sits inside the false bulk for every solver, the real-geometry form of the same efficiency ceiling.